Probabilistic Irreducibility Tests in $\mathbb{F}[T]$

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1 Introduction

Given a positive integer N, it is often natural and intuitive to ask: is N prime? This is because primes have intriguing and invaluable properties in number theory and other areas of mathematics. However, the factorization of large natural numbers into primes is such a computationally difficult task that the widely used RSA cryptographic algorithm banks on how difficult it is.

Therefore, deciding the primality of a large number is an important problem. Fortunately, three primality tests can help us determine if a number is prime or composite without having to factor it. These tests are the Fermat test, the Solovay–Strassen test, and the Miller–Rabin test. Through trial and error, each test can identify "witnesses" that break a property of prime numbers, thus proving compositeness. The second and third tests are guaranteed to have a high proportion of witnesses for any composite number they are applied to, so they can both be considered probabilistic primality tests. The first test is not always guaranteed to have a high proportion of witnesses for composite numbers, which is what motivated the discovery of the second and third tests.

This paper explores analogues of these three tests to determine reducibility or irreducibility of polynomials in $\mathbb{F}_p[T]$. Determining if polynomials are irreducible plays an important role in number theory, cryptography, and Galois theory. To formulate and prove the polynomial versions of these tests and their properties, we extend key results in number theory from $\mathbb{Z}$ to $\mathbb{F}_p[T]$, such as Fermat's little theorem, Euler's congruence, and quadratic/Jacobi reciprocity.

2 The Fermat Test

2.1 The Fermat Test in $\mathbb{Z}$

The Fermat test in $\mathbb{Z}$ is one method of proving an integer is composite *without* factoring it. The test is based on breaking the congruence appearing in Fermat's little theorem. Here is the basic idea.

Theorem 2.1. *Let $n > 1$ be in $\mathbb{Z}$. If there exists $a \not\equiv 0 \bmod n$ such that*

$$a^{n-1} \not\equiv 1 \bmod n,$$

then n is necessarily composite.

Proof. Fermat's little theorem says that if n is prime then $a^{n-1} \equiv 1 \bmod n$ for every $a \not\equiv 0 \bmod n$. Now look at the contrapositive: if $a^{n-1} \not\equiv 1 \bmod n$ for some $a \not\equiv 0 \bmod n$, then n is not prime, so if $n > 1$ then n has to be composite. □

Definition 2.2. For $n > 1$, an a such that $1 \leq a \leq n-1$ and $a^{n-1} \not\equiv 1 \bmod n$ is called a *Fermat witness* for n.

Example 2.3. Let $n = 5461$. We have $2^{n-1} = 2^{5460} \equiv 1 \bmod 5461$, which is consistent with the possibility that 5461 is prime. But $3^{5460} \equiv 1377 \bmod 5461$, which is inconsistent with 5461 being prime. Thus 5461 is composite and 3 is a Fermat witness for 5461 while 2 is not.

In fact $5461 = 43 \cdot 127$, but the way we saw 5461 is composite using the number 3 did not reveal how 5461 factors.

Example 2.4. Let $n = 721801$. The table below shows 721801 has 11 as its first Fermat witness.

a	2	3	4	5	6	7	8	9	10	11
$a^{721800} \bmod 721801$	1	1	1	1	1	1	1	1	1	719398

We thus know 721801 is composite without knowing a nontrivial factorization. (It is $601 \cdot 1201$.)

If $a^{n-1} \equiv 1 \bmod n$ and $b^{n-1} \equiv 1 \bmod n$, then $(ab)^{n-1} \equiv 1 \bmod n$, so the first Fermat witness for a composite number n will always be prime, as we saw in the previous two examples.

Remark 2.5. We restrict Fermat witnesses for n, if there are any, to be in the range $\{1, \ldots, n-1\}$ since the value of $a^{n-1} \bmod n$ depends on a only through its remainder modulo n.

If n is prime then it has no Fermat witness by Fermat's little theorem. If n is composite then it has a Fermat witness, such as any factor of n between 1 and n: if $d \mid n$ and $1 < d < n$ then $d^{n-1} \not\equiv 1 \bmod n$ since otherwise d would be a unit mod n, which is not the case. More generally, any a in $\{1, \ldots, n-1\}$ with $\gcd(a, n) > 1$ is a Fermat witness for n. Most composite n also have Fermat witnesses that are relatively prime to n, which we saw in the previous two examples: $(3, 5461) = 1$ and $(11, 721801) = 1$. This is significant since it can let us detect that n is composite using numbers having "nothing in common" with n. Unfortunately, some composite n have no such Fermat witness:

Definition 2.6. A composite number n is called a *Carmichael number* if for every Fermat witnesses a of n, $\gcd(a, n) > 1$. That is, $a^{n-1} \not\equiv 1 \bmod n$ only when $\gcd(a, n) > 1$. Equivalently, $\gcd(a, n) = 1 \implies a^{n-1} \equiv 1 \bmod n$.

Example 2.7. The number 6601 is a Carmichael number: it is $7 \cdot 23 \cdot 41$ and $a^{6600} \not\equiv 1 \bmod 6601$ only when $\gcd(a, 6601) \neq 1$. Thus whenever $\gcd(a, 6601) = 1$ we have $a^{6600} \equiv 1 \bmod 6601$. The only way to break that congruence is to use an a that has a factor in common with 6601.

The following theorem gives an alternate description of Carmichael numbers.

Theorem 2.8 (Korselt's Criterion)**.** *A composite integer n is Carmichael if and only if the following holds: (1) n is squarefree, and (2) for every prime p dividing n, $p - 1 \mid n - 1$.*

Proof. See [1] for a proof. □

Fermat witnesses of Carmichael numbers sometimes can be rare, so we seek primality tests where composite numbers are assured to have many witnesses, or in other words where there is nothing like Carmichael numbers for those tests.

2.2 The Fermat Test in $\mathbb{F}_p[T]$

The polynomial ring $\mathbb{F}_p[T]$ admits a test like the Fermat test from $\mathbb{Z}$ that can distinguish between reducible and irreducible polynomials without factoring.

Definition 2.9. For nonzero f in $\mathbb{F}_p[T]$, we write $\mathbb{F}_p[T]_f$ or $\mathbb{F}_p[T]/(f)$ for the ring of polynomials in $\mathbb{F}_p[T]$ with addition and multiplication taken modulo f.

Definition 2.10. Define the *norm* of a nonzero $f \in \mathbb{F}_p[T]$ to be $\mathrm{N}(f) = p^{\deg(f)}$.

Here is an important combinatorial interpretation of $\mathrm{N}(f)$.

Proposition 2.11. *For nonzero f in $\mathbb{F}_p[T]$, the ring $\mathbb{F}_p[T]_f$ has size $\mathrm{N}(f)$.*

Proof. Let $d = \deg \pi$. By the division algorithm in $\mathbb{F}_p[T]$ every $g \in \mathbb{F}_p[T]$ can be written uniquely as $fq + r$ where q and r are in $\mathbb{F}_p[T]$ and $r = 0$ or $\deg r < d$. Thus, the elements in $\mathbb{F}_p[T]_f$ can be uniquely represented by an r such that

$$r = c_0 + c_1 T + \ldots + c_{d-1} T^{d-1},$$

where each c_i is in $\mathbb{F}_p$. The number of such polynomials r is $p^{\deg f} = \mathrm{N}(f)$. □

The formula $|\mathbb{F}_p[T]_f| = \mathrm{N}(f)$ when f is nonzero is analogous to the formula $|\mathbb{Z}_n| = |n|$ for an nonzero integer n. Note that both formulae are multiplicative: we have $|mn| = |m||n|$ and $\mathrm{N}(fg) = \mathrm{N}(f)\mathrm{N}(g)$.

Using Proposition 2.11, we can formulate the analogue of Fermat's little theorem in $\mathbb{F}_p[T]$.

Theorem 2.12. *Let π be irreducible in $\mathbb{F}_p[T]$. Then for each $a \in \mathbb{F}_p[T]$ such that $a \not\equiv 0 \bmod \pi$,*

$$a^{\mathrm{N}(\pi)-1} \equiv 1 \bmod \pi.$$

Proof. We claim that when π is irreducible, the nonzero elements of $\mathbb{F}_p[T]_\pi$ equipped with multiplication modulo π forms a group, denoted $\mathbb{F}_p[T]_\pi^\times$. The existence of an identity and associativity of multiplication is easily verifiable, and the existence of inverses follows from the Euclidean algorithm in $\mathbb{F}_p[T]$. Since we exclude 0, $\mathbb{F}_p[T]_\pi^\times$ is a group of order $\mathrm{N}(\pi)-1$. By Lagrange's theorem for the group $\mathbb{F}_p[T]_\pi^\times$, when $a \not\equiv 0 \bmod \pi$ we get

$$a^{\mathrm{N}(\pi)-1} \equiv 1 \bmod \pi. \qquad \square$$

Using Theorem 2.12, Fermat's test in $\mathbb{Z}$ has the following immediate analogue in $\mathbb{F}_p[T]$.

Theorem 2.13. *Let $f \in \mathbb{F}_p[T]$ be nonconstant. If there exists an $a \in \mathbb{F}_p[T]$ such that $a \not\equiv 0 \bmod f$ and*

$$a^{\mathrm{N}(f)-1} \not\equiv 1 \bmod f,$$

then f is reducible.

A search for for a such that $a^{\mathrm{N}(f)-1} \not\equiv 1 \bmod f$ only needs to be conducted in the range

$$\{a \in \mathbb{F}_p[T] \colon \deg a < \deg f\}$$

for the same reason we use $\{1, \dots, n-1\}$ when searching for Fermat witnesses for n in $\mathbb{Z}$: $a^{\mathrm{N}(f)-1} \bmod f$ depends on a only through its remainder modulo f.

Definition 2.14. For nonconstant $f \in \mathbb{F}_p[T]$, an $a \in \mathbb{F}_p[T]$ such that $\deg a < \deg f$ is called a *Fermat witness* for f if

$$a^{\mathrm{N}(f)-1} \not\equiv 1 \bmod f.$$

Otherwise, we say a is a *Fermat nonwitness*.

Example 2.15. In $\mathbb{F}_3[T]$, let $f(T) = T^5 + T^2 + 2$, so $\mathrm{N}(f) = 3^5 = 243$. When $a = T$, $a^{\mathrm{N}(f)-1} = T^{242} \equiv T + 1 \not\equiv 1 \bmod f$. Thus T is a Fermat witness for f, which shows f is reducible in $\mathbb{F}_3[T]$. Note that factoring was not needed to show reducibility.

Irreducible f have no Fermat witness by Theorem 2.12. Reducible f are *guaranteed* to have a Fermat witness: use any a in $\mathbb{F}_p[T]$ with $\deg a < \deg f$ where $\gcd(a, f) \neq 1$, such as any factor of f with degree strictly between 0 and $\deg f$. As in $\mathbb{Z}$, most reducible f have Fermat witnesses that are relatively prime to f. However, there exist reducible f whose Fermat witnesses are never relatively prime to f, analogous to Carmichael numbers in $\mathbb{Z}$.

Definition 2.16. A reducible $f \in \mathbb{F}_p[T]$ is said to be *Carmichael* if none of its Fermat witnesses are relatively prime to f. Alternatively, a reducible f is Carmichael if the following holds:

$$\gcd(a, f) = 1 \implies a^{\mathrm{N}(f)-1} \equiv 1 \bmod f.$$

An equivalent condition for f to be Carmichael is provided in Section 2.4.

2.3 Proportion of Fermat Witnesses

Primes in $\mathbb{Z}$ and irreducibles in $\mathbb{F}_p[T]$ have no Fermat witnesses while a composite positive integer or reducible polynomial in $\mathbb{F}_p[T]$ always has at least one Fermat witness. If we want to test primality or irreducibility, it would be reassuring to know composite integers and reducible polynomials have a substantial proportion of Fermat witnesses. We'd then know in advance that finding a Fermat witness should not take too many applications of the Fermat test when such witnesses exist.

Theorem 2.17. *For $n \geq 2$, if some integer b satisfies $b^{n-1} \not\equiv 1 \bmod n$ and $(b, n) = 1$, then $|\{1 \leq a \leq n-1 : a^{n-1} \not\equiv 1 \bmod n\}| > (n-1)/2$.*

Proof. Define the sets A, B, and C as follows:

$$A = \{1 \leq a \leq n-1 : a^{n-1} \equiv 1 \bmod n\}$$
$$B = \{1 \leq a \leq n-1 : (a, n) = 1 \text{ and } a^{n-1} \not\equiv 1 \bmod n\}$$
$$C = \{1 \leq a \leq n-1 : (a, n) > 1\}.$$

We want to show $|B \cup C| > (n-1)/2$.

Claim: the set A with multiplication taken modulo n is a subgroup of $\mathbb{Z}_n^\times$.

To prove this, note that

$$A \cup B = \{1 \leq a \leq n-1 : (a, n) = 1\} = \mathbb{Z}_n^\times$$

and $A \cup B = \mathbb{Z}_n^\times$ is a group. Therefore, we know $A \subset \mathbb{Z}_n^\times$.

Let a, b be fixed and arbitrary elements in A. We want to show ab^{-1} is in A. We have

$$(ab^{-1})^{n-1} = a^{n-1}b^{-(n-1)} \equiv 1 \bmod n$$

because a, b are in A. Also, since $(a, n) = 1$ and $(b, n) = 1$, $(ab^{-1}, n) = 1$. So ab^{-1} is in A and A is a subgroup of $\mathbb{Z}_n^\times$. That proves the claim.

Since B is not empty, A is a proper subgroup of $A \cup B = \mathbb{Z}_n^\times$, so Lagrange's theorem implies that $|A \cup B| \geq 2|A|$, since $|A|$ divides $|A \cup B|$. Thus $|A| \leq \varphi(n)/2$. Since n has a Fermat witness, it is composite, so $\varphi(n) < n-1$. Thus $|A| < (n-1)/2$. Since $|A \cup B \cup C| = n-1$ and A, B, and C are disjoint, that $|A| < (n-1)/2$ implies $|B \cup C| > (n-1)/2$. □

Theorem 2.17 generalizes to $\mathbb{F}_p[T]$.

Theorem 2.18. *For nonconstant $\varphi \in \mathbb{F}_p[T]$, if some g in $\mathbb{F}_p[T]$ satisfies $g^{\mathrm{N}(\varphi)-1} \not\equiv 1 \bmod \varphi$ and $(g, \varphi) = 1$ then*

$$|\{f \mid 1 \leq \deg(f) \leq \deg(\varphi) - 1 \text{ and } f^{\mathrm{N}(\varphi)-1} \not\equiv 1 \bmod \varphi\}| > \frac{\mathrm{N}(\varphi) - 1}{2}.$$

Proof. Let

$$A = \{f \mid 0 \leq \deg(f) \leq \deg(\varphi) - 1 \text{ and } f^{\mathrm{N}(\varphi)-1} \equiv 1 \bmod \varphi\}$$
$$B = \{f \mid 0 \leq \deg(f) \leq \deg(\varphi) - 1 \text{ and } (f, \varphi) = 1 \text{ and } f^{\mathrm{N}(\varphi)-1} \not\equiv 1 \bmod \varphi\}$$
$$C = \{f \mid 0 \leq \deg(f) \leq \deg(\varphi) - 1 \text{ and } (f, \varphi) \neq 1\}.$$

Claim: the set A with multiplication taken modulo φ is a subgroup of $(\mathbb{F}_p[T]_\varphi)^\times$.

Since

$$A \cup B = \{f \mid 0 \leq \deg(f) \leq \deg(\varphi) - 1 \text{ and } (f, \varphi) = 1\} = (\mathbb{F}_p[T]_\varphi)^\times,$$

we have $A \subset (\mathbb{F}_p[T]_\varphi)^\times$. Let α and β be fixed and arbitrary elements in A. We want to show that $\alpha\beta^{-1}$ is in A. We have

$$(\alpha\beta^{-1})^{\mathrm{N}(\varphi)-1} = \alpha^{\mathrm{N}(\varphi)-1}\beta^{-(\mathrm{N}(\varphi)-1)} \equiv 1 \bmod \varphi$$

since α, β are in A. Also, since $(\alpha, \varphi) = 1$ and $(\beta, \varphi) = 1$, $(\alpha\beta^{-1}, \varphi) = 1$. So $\alpha\beta^{-1}$ is in A and, as a result, A is a subgroup of $(\mathbb{F}_p[T]_\varphi)^\times$. That proves the claim.

Since B is not empty, A is a proper subgroup of $A \cup B = (\mathbb{F}_p[T]_\varphi)^\times$, so Lagrange's theorem implies that $|A \cup B| \geq 2|A|$, since $|A|$ divides $|A \cup B|$. Thus $|A| \leq |(\mathbb{F}_p[T]_\varphi)^\times|/2$. Since φ has a Fermat witness, it is reducible, so $|(\mathbb{F}_p[T]_\varphi)^\times| < \mathrm{N}(\varphi) - 1$. Thus $|A| < (\mathrm{N}(\varphi) - 1)/2$. We know that $|A \cup B \cup C| = \mathrm{N}(\varphi) - 1$ and A, B, and C are disjoint, so from $|A| < (\mathrm{N}(\varphi) - 1)/2$ we get

$$|B \cup C| > \frac{\mathrm{N}(\varphi) - 1}{2}.$$ □

2.4 Korselt's Criterion in $\mathbb{F}_p[T]$

Here, we present the analogue of Korselt's criterion for Carmichael polynomials. Recall Korselt's criterion for Carmichael numbers in Theorem 2.8.

Theorem 2.19 (Korselt's Criterion in $\mathbb{F}_p[T]$)**.** *A nonconstant reducible f in $\mathbb{F}_p[T]$ is Carmichael if and only if (1) f is squarefree (it has no nonconstant square factor) and (2) for every irreducible π dividing f, $\mathrm{N}(\pi) - 1 \mid \mathrm{N}(f) - 1$. Condition (2) is equivalent to $\deg \pi \mid \deg f$ for all irreducible factors π of f.*

Proof. Assume that f is Carmichael. We will first show that f is squarefree. Suppose f is not squarefree—i.e., there exists an irreducible π such that $f = \pi^k g$ for some $g \in \mathbb{F}_p[T]$ satisfying $(\pi, g) = 1$ and $k \geq 2$. By the Chinese remainder theorem in $\mathbb{F}_p[T]$, we can choose $a \in \mathbb{F}_p[T]$ such that $a \equiv 1 + \pi \bmod \pi^k$ and $a \equiv 1 \bmod g$. Since a is coprime to both π^k and g, we have $(a, f) = 1$. Therefore, by the definition of Carmichael polynomials, we get

$$a^{\mathrm{N}(f)-1} \equiv 1 \bmod f.$$

Reducing this congruence modulo π^2, we get

$$a^{\mathrm{N}(f)-1} \equiv 1 \bmod \pi^2.$$

We have $a \equiv 1 + \pi \bmod \pi^2$ so $(1+\pi)^{\mathrm{N}(f)-1} \equiv 1 \bmod \pi^2$. Using the binomial theorem,

$$(1+\pi)^{\mathrm{N}(f)-1} \equiv 1 + (\mathrm{N}(f)-1)\pi \bmod \pi^2.$$

As a coefficient, $\mathrm{N}(f) = p^{\deg f} = 0$ since $p = 0$ in $\mathbb{F}_p$. Therefore,

$$1 \equiv (1+\pi)^{\mathrm{N}(f)-1} \equiv 1 + \cdot(\mathrm{N}(f)-1)\pi \equiv 1 - \pi \bmod \pi^2,$$

which implies $0 \equiv \pi \bmod \pi^2$, which is a contradiction. Therefore f is squarefree, which is condition (1).

Next, we show that $\mathrm{N}(\pi) - 1 \mid \mathrm{N}(f) - 1$ for all irreducible divisors π of f. Choose an irreducible π dividing f. Since f is squarefree, $(\pi, f/\pi) = 1$. The finite field $\mathbb{F}_p[T]_\pi$ has order $\mathrm{N}(\pi)$, and just as $\mathbb{F}_p^\times$ is a cyclic group under multiplication, it can be shown that $\mathbb{F}_p[T]_\pi^\times$ is cyclic [2], so there is $g \in \mathbb{F}_p[T]_\pi$ such that g generates $\mathbb{F}_p[T]_\pi^\times$, which means the order of g in $\mathbb{F}_p[T]_\pi^\times$ is $\mathrm{N}(\pi)-1$. Now by the Chinese remainder theorem there is h such that $h \equiv g \bmod \pi$ and $h \equiv 1 \bmod f/\pi$. In particular, $(h, f) = 1$. Since f is Carmichael, $h^{\mathrm{N}(f)-1} \equiv 1 \bmod f$. Reducing modulo π, $g^{\mathrm{N}(f)-1} \equiv 1 \bmod \pi$. Thus $\mathrm{N}(\pi) - 1 \mid \mathrm{N}(f) - 1$ since the order of $g \bmod \pi$ is $\mathrm{N}(\pi) - 1$. We have proved condition (2).

Now assume conditions (1) and (2) hold. We will show f is Carmichael. By (1), $f = c\pi_1 \cdots \pi_k$ where c is a nonzero constant and the π_i's are distinct monic irreducibles. Let $(a, f) = 1$. By Fermat's little theorem, $a^{\mathrm{N}(\pi_i)-1} \equiv 1 \bmod \pi_i$ for each i. Since $\mathrm{N}(\pi_i) - 1 \mid \mathrm{N}(f) - 1$ by condition (2),

$$a^{\mathrm{N}(f)-1} \equiv 1 \bmod \pi_i$$

for each i. Since the π_i's are pairwise relatively prime, those congruences mod each π_i can be combined into the single congruence

$$a^{\mathrm{N}(f)-1} \equiv 1 \bmod f.$$

This holds for all a such that $(a, f) = 1$ and f is reducible, so f is Carmichael. □

3 The Solovay–Strassen Test

A second primality test in $\mathbb{Z}$, introduced by Solovay and Strassen [7], [8] is based on breaking Euler's congruence for the Legendre symbol rather than the congruence in Fermat's little theorem. We will review the Solovay–Strassen test in $\mathbb{Z}$ and then present analogues of it in $\mathbb{F}_p[T]$ when $p > 2$ and then when $p = 2$.

3.1 Solovay–Strassen in $\mathbb{Z}$

Let's recall the Legendre symbol and some of its properties.

Definition 3.1. For an odd prime p and $a \in \mathbb{Z}$,

$$\left(\frac{a}{p}\right) = \begin{cases} 1, \text{ if } a \equiv \square \bmod p \text{ and } a \not\equiv 0 \bmod p, \\ 0, \text{ if } a \equiv 0 \bmod p, \\ -1, \text{ if } a \not\equiv \square \bmod p. \end{cases}$$

where $\square$ means a square.

Here are some properties of the Legendre symbol on $\mathbb{Z}$, where a and b are any integers:

1. if $a \equiv b \bmod p$, then $(\frac{a}{p}) = (\frac{b}{p})$,
2. $a^{\frac{p-1}{2}} \equiv (\frac{a}{p}) \bmod p$,
3. $(\frac{ab}{p}) = (\frac{a}{p})(\frac{b}{p})$,
4. $(\frac{-1}{p}) = (-1)^{(p-1)/2}$,
5. $(\frac{2}{p}) = (-1)^{(p^2-1)/8}$.
6. For odd primes p and q, $(\frac{q}{p}) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}(\frac{p}{q})$.

The last three properties are the quadratic reciprocity law in $\mathbb{Z}$ (the two supplementary laws and the main law). They have the equivalent forms

$$\left(\frac{-1}{p}\right) = \begin{cases} 1, & \text{if } p \equiv 1 \bmod 4, \\ -1, & \text{if } p \equiv 3 \bmod 4, \end{cases} \quad \left(\frac{2}{p}\right) = \begin{cases} 1, & \text{if } p \equiv 1 \text{ or } 7 \bmod 8, \\ -1, & \text{if } p \equiv 3 \text{ or } 5 \bmod 8, \end{cases}$$

$$\left(\frac{q}{p}\right) = \begin{cases} (\frac{p}{q}), & \text{if } p \text{ or } q \text{ is } 1 \bmod 4, \\ -(\frac{p}{q}), & \text{if } p \text{ and } q \text{ are } 3 \bmod 4. \end{cases}$$

We will not prove any of the above Legendre symbol properties, as their proofs are not the focus of this paper. Instead we will be concerned with *applications* of these properties.

An extension of the Legendre symbol that allows composite odd denominators is the Jacobi symbol.

Definition 3.2. For odd $n > 1$ in $\mathbb{Z}$, write $n = p_1 p_2 \dots p_k$, where the p_i's are odd primes that are not necessarily distinct. For each $a \in \mathbb{Z}$, set the *Jacobi symbol* of a and n to be

$$\left(\frac{a}{n}\right) = \left(\frac{a}{p_1}\right) \dots \left(\frac{a}{p_k}\right).$$

When $n = 1$, set $(\frac{a}{1}) = 1$ when a is nonzero and $(\frac{0}{1}) = 0$.

If n is an odd prime, then the Jacobi symbol $(\frac{a}{n})$ is a Legendre symbol. If $(a,n) > 1$, then a prime p_i dividing n also divides a, making $(\frac{a}{p_i}) = 0$, so $(\frac{a}{n}) = 0$. If $(a,n) = 1$, then $(\frac{a}{p_i}) = \pm 1$ for all primes p_i dividing n, so $(\frac{a}{n}) = \pm 1$.

If $(\frac{a}{n}) = -1$, then a is not a quadratic residue mod n: $(\frac{a}{n}) = -1 \implies (\frac{a}{p_i}) = -1$ for some $p_i \mid n$, and if it were the case that $a \equiv x^2 \bmod n$, then $(x,n) = 1$ and $a \equiv x^2 \bmod p_i$ with $x \not\equiv 0 \bmod p_i$, which contradicts $(\frac{a}{p_i})$ being -1. Thus $a \not\equiv \square \bmod n$. However, it is *not* necessarily true that if $(\frac{a}{n}) = 1$, then a is a quadratic residue mod n. For example, $(\frac{2}{15}) = (\frac{2}{3})(\frac{2}{5}) = (-1)(-1) = 1$, but 2 is not a quadratic residue mod 15 since it is not a quadratic residue mod 3 (or mod 5).

The Jacobi symbol satisfies properties similar to those of the Legendre symbol aside from $a^{\frac{p-1}{2}} \equiv (\frac{a}{p}) \bmod p$: for any integers a and b and any odd positive integers m and n,

1. if $a \equiv b \bmod n$, then $(\frac{a}{n}) = (\frac{b}{n})$,
2. $(\frac{ab}{n}) = (\frac{a}{n})(\frac{b}{n})$ and $(\frac{a}{mn}) = (\frac{a}{m})(\frac{a}{n})$,
3. $(\frac{-1}{n}) = (-1)^{(n-1)/2}$,
4. $(\frac{2}{n}) = (-1)^{(n^2-1)/8}$,
5. $(\frac{n}{m}) = (-1)^{\frac{m-1}{2}\cdot\frac{n-1}{2}}(\frac{m}{n})$.

The last three properties are called *Jacobi reciprocity* (the two supplementary laws and the main law) and they have the equivalent forms

$$\left(\frac{-1}{n}\right) = \begin{cases} 1, & \text{if } n \equiv 1 \bmod 4, \\ -1, & \text{if } n \equiv 3 \bmod 4, \end{cases} \quad \left(\frac{2}{n}\right) = \begin{cases} 1, & \text{if } n \equiv 1 \text{ or } 7 \bmod 8, \\ -1, & \text{if } n \equiv 3 \text{ or } 5 \bmod 8, \end{cases}$$

$$\left(\frac{n}{m}\right) = \begin{cases} (\frac{m}{n}), & \text{if } m \text{ or } n \text{ is } 1 \bmod 4, \\ -(\frac{m}{n}), & \text{if } m \text{ and } n \text{ are } 3 \bmod 4, \end{cases}$$

which are better from a programming perspective.

Jacobi reciprocity allows us to compute any Jacobi symbol $(\frac{a}{n})$ in a computationally efficient manner without factoring the numerator or denominator other than stripping out factors of -1 and 2 from the numerator and using properties (3) and (4). Then simplify the Jacobi symbol by flipping the two entries using property (5), reduce the numerator modulo the denominator using property (1), and repeat these steps again. Computing the Jacobi symbol in $\mathbb{Z}$, therefore, is of comparable complexity to computing the gcd in $\mathbb{Z}$ via the Euclidean algorithm.

Remark 3.3. The main law of Jacobi reciprocity, property (5), is often stated for odd positive *relatively prime* m and n, but it is true when m and n are not relatively prime since the main law in that case says $0 = 0$. If we start

calculating a Jacobi symbol $(\frac{a}{n})$ where a and n are not relatively prime, then this fact will be revealed when we calculate that $(\frac{a}{n}) = 0$. For example,

$$\left(\frac{57}{323}\right) = \left(\frac{323}{57}\right) = \left(\frac{38}{57}\right) = \left(\frac{2}{57}\right)\left(\frac{19}{57}\right) = 1 \cdot \left(\frac{57}{19}\right) = \left(\frac{0}{19}\right) = 0,$$

which implies $(57, 323) > 1$, In fact, $(57, 323) = 19$.

The second property of the Legendre symbol listed above is called Euler's congruence. It says that for odd primes p and all $a \in \mathbb{Z}$, $a^{(p-1)/2} \equiv (\frac{a}{p}) \bmod p$. This has mainly theoretical rather than computational importance, being used to prove some of the other properties we listed for the Legendre symbol. Most properties of the Jacobi symbol are derived from analogous properties of the Legendre symbol, bypassing a Jacobi symbol analogue of Euler's congruence.

We can still ask whether or not for each odd composite n the congruence $a^{(n-1)/2} \equiv (\frac{a}{n}) \bmod n$ with the Jacobi symbol on the right side is true for all $a \bmod n$. This turns out to fail *a lot*, which leads to the Solovay–Strassen test.

Theorem 3.4. *For odd $n > 1$ in $\mathbb{Z}$, if $a^{(n-1)/2} \not\equiv (\frac{a}{n}) \bmod n$ for some $a \not\equiv 0 \bmod n$, then n is composite.*

An $a \in \{1, \ldots, n-1\}$ such that either $(\frac{a}{n}) = 0$ or $a^{(n-1)/2} \not\equiv (\frac{a}{n}) \bmod n$ if $(a, n) = 1$ is called an *Euler witness* for n since it breaks Euler's congruence.

Example 3.5. Let $n = 5461$. In Example 2.3 we saw that $2^{n-1} \equiv 1 \bmod n$, so 2 is not a Fermat witness for n. But it is an Euler witness: $2^{(n-1)/2} = 2^{2730} \equiv 1 \bmod n$ while $(\frac{2}{n}) = -1$ since $n \equiv 5 \bmod 8$, so $2^{(n-1)/2} \not\equiv (\frac{2}{n}) \bmod n$.

Example 3.6. Let $n = 721801$. In Example 2.4 we saw the first Fermat witness for n is 11. Its first Euler witness is 5, as indicated in the table below.

a	2	3	4	5
$a^{(n-1)/2} \bmod n$	1	1	1	719398
$(\frac{a}{n}) \bmod n$	1	1	1	721800

An odd prime n has no Euler witness. Solovay and Strassen [7], [8] showed that for any odd composite $n > 1$, the proportion of $a \in \{1, \ldots, n-1\}$ that are Euler witnesses for n is greater than 50%, or equivalently the proportion of Euler nonwitnesses for n is less than 50%.

Legendre and Jacobi symbols can be defined for polynomials with coefficients in a finite field and they satisfy a generalization of quadratic reciprocity and Jacobi reciprocity. This leads to a polynomial Solovay–Strassen test.

3.2 Extension of Legendre/Jacobi Symbols to $\mathbb{F}_p[T]$

Recall the definition of $\mathrm{N}(f)$ as $p^{\deg f}$. Instead of $a^{(n-1)/2}$ for $a \in \mathbb{Z}_n$ when $n > 1$ is odd, we will use the polynomial analogue $a^{(\mathrm{N}(f)-1)/2}$ for $a \in \mathbb{F}_p[T]_f$ when f is nonconstant and $p > 2$. The analogue when $p = 2$ is in Section 3.4.

We naturally extend the notion of the Legendre symbol to polynomials in $\mathbb{F}_p[T]$ when $p > 2$ as follows.

Definition 3.7. For prime $p > 2$, irreducible $\pi \in \mathbb{F}_p[T]$, and $a \in \mathbb{F}_p[T]$, let

$$\left(\frac{a}{\pi}\right) = \begin{cases} 1, & \text{if } a \equiv \square \bmod \pi \text{ and } a \not\equiv 0 \bmod \pi, \\ 0, & \text{if } a \equiv 0 \bmod \pi, \\ -1, & \text{if } a \not\equiv \square \bmod \pi. \end{cases}.$$

The Legendre symbol satisfies properties like the Legendre symbol in $\mathbb{Z}$. Here are the first three.

1. $f_1 \equiv f_2 \bmod \pi \implies (\frac{f_1}{\pi}) = (\frac{f_2}{\pi})$,
2. (Euler's congruence) $f^{(\mathrm{N}(\pi)-1)/2} \equiv (\frac{f}{\pi}) \bmod \pi$,
3. $(\frac{fg}{\pi}) = (\frac{f}{\pi})(\frac{g}{\pi})$.

Proof. The first property is true by definition.

To prove Euler's congruence, consider the polynomial $P(x) = x^{\frac{\mathrm{N}(\pi)-1}{2}} - 1$. Since $\mathbb{F}_p[T]_\pi$ is a field, $P(x)$ has at most $\frac{\mathrm{N}(\pi)-1}{2}$ roots in that field. If $a \in \mathbb{F}_p[T]$ is a nonzero square modulo π, then $a \equiv b^2 \bmod \pi$ for some $b \not\equiv 0 \bmod \pi$, so $a^{\frac{\mathrm{N}(\pi)-1}{2}} \equiv b^{\mathrm{N}(\pi)-1} \equiv 1 \bmod \pi$ by Theorem 2.12. Thus $a \bmod \pi$ is a root of $P(x)$. This shows that $(\frac{a}{\pi}) = 1 \implies P(a) \equiv 0 \bmod \pi$.

Since π is irreducible, $a^2 \equiv b^2 \bmod \pi \iff \pi \mid (a^2 - b^2) \iff \pi \mid a \pm b$, so $a^2 \equiv b^2 \bmod \pi$ if and only if $a \equiv b \bmod \pi$ or $a \equiv -b \bmod \pi$. Thus, exactly half the nonzero elements of $\mathbb{F}_p[T]_\pi$ are squares. By the previous paragraph these nonzero squares are roots of P in $\mathbb{F}_p[T]_\pi$. Since P has degree $(\mathrm{N}(\pi) - 1)/2$, it has at most $(\mathrm{N}(\pi) - 1)/2$ roots in $\mathbb{F}_p[T]_\pi$. We already found that many roots in $\mathbb{F}_p[T]_\pi$ by using the nonzero squares mod π, so those are the only roots of P in $\mathbb{F}_p[T]_\pi$. Thus if $(\frac{a}{\pi}) = -1$, then $P(a) \not\equiv 0 \bmod \pi$: $a^{\frac{\mathrm{N}(\pi)-1}{2}} \not\equiv 1 \bmod \pi$. Since $a \not\equiv 0 \bmod \pi$, $(a^{\frac{\mathrm{N}(\pi)-1}{2}})^2 \equiv a^{\mathrm{N}(\pi)-1} \equiv 1 \bmod \pi$ so $a^{\frac{\mathrm{N}(\pi)-1}{2}} \equiv -1 \equiv (\frac{a}{\pi}) \bmod \pi$. Lastly, the congruence $a^{\frac{\mathrm{N}(\pi)-1}{2}} \equiv (\frac{a}{\pi}) \bmod \pi$ is clear when $a \equiv 0 \bmod \pi$.

Now, we show property 3, and will use Euler's congruence. By basic algebra $(fg)^{\frac{\mathrm{N}(\pi)-1}{2}} \equiv f^{\frac{\mathrm{N}(\pi)-1}{2}} g^{\frac{\mathrm{N}(\pi)-1}{2}} \bmod \pi$, so by Euler's congruence $(\frac{fg}{\pi}) \equiv (\frac{f}{\pi})(\frac{g}{\pi}) \bmod \pi$. Since $(\frac{fg}{\pi})$ and $(\frac{f}{\pi})(\frac{g}{\pi})$ are both among 0, 1, or -1, which are never congruent mod π since $p > 2$, that $(\frac{fg}{\pi}) \equiv (\frac{f}{\pi})(\frac{g}{\pi}) \bmod \pi$ implies $(\frac{fg}{\pi}) = (\frac{f}{\pi})(\frac{g}{\pi})$. □

The Legendre symbol on $\mathbb{F}_p[T]$ has one more vital property: it satisfies a quadratic reciprocity law. In $\mathbb{Z}$ there is a main law applying to odd primes and supplementary laws for $(\frac{-1}{p})$ and $(\frac{2}{p})$. Similarly, a quadratic reciprocity law in $\mathbb{F}_p[T]$ has multiple parts: a main law for monic irreducibles and a supplementary law for nonzero constants.

Theorem 3.8. *For prime $p > 2$ and monic irreducibles $\pi_1, \pi_2 \in \mathbb{F}_p[T]$,*

$$\left(\frac{\pi_2}{\pi_1}\right) = (-1)^{\frac{\mathrm{N}(\pi_1)-1}{2} \cdot \frac{\mathrm{N}(\pi_2)-1}{2}} \left(\frac{\pi_1}{\pi_2}\right).$$

For nonzero $c \in \mathbb{F}_p$ and monic irreducible $\pi \in \mathbb{F}_p[T]$,

$$\left(\frac{c}{\pi}\right) = c^{(\mathrm{N}(\pi)-1)/2}.$$

Proof. The first part is simple when $\pi_1 = \pi_2$ since the equation in that case becomes $0 = 0$. For the rest, we refer to the proof in [4]. □

Remark 3.9. The formulas can be rewritten as $(\frac{\pi_2}{\pi_1}) = (-1)^{\frac{p-1}{2}(\deg \pi_1)(\deg \pi_2)}(\frac{\pi_1}{\pi_2})$ and $(\frac{c}{\pi}) = (\frac{c}{p})^{\deg \pi}$. To see why, since p is odd for any $d \geq 1$ we have

$$\frac{p^d-1}{2} = \frac{(p-1)(1+p+\cdots+p^{d-1})}{2} = \frac{p-1}{2}(1+p+\cdots+p^{d-1}) \equiv \frac{p-1}{2}d \bmod 2.$$

Thus $\frac{\mathrm{N}(f)-1}{2} \equiv \frac{p-1}{2} \deg f \bmod 2$ for nonzero f, so

$$\begin{aligned}\frac{\mathrm{N}(\pi_1)-1}{2} \cdot \frac{\mathrm{N}(\pi_2)-1}{2} &\equiv \left(\frac{p-1}{2}\right)^2 \cdot \deg \pi_1 \cdot \deg \pi_2 \bmod 2 \\ &\equiv \left(\frac{p-1}{2}\right) \cdot \deg \pi_1 \cdot \deg \pi_2 \bmod 2\end{aligned}$$

since $k^2 \equiv k \bmod 2$ for all k, so $(-1)^{\frac{\mathrm{N}(\pi_1)-1}{2}\frac{\mathrm{N}(\pi_2)-1}{2}} = (-1)^{\frac{p-1}{2}(\deg \pi_1)(\deg \pi_2)}$. In particular, if $p \equiv 1 \bmod 4$, then the main law says $(\frac{\pi_2}{\pi_1}) = (\frac{\pi_1}{\pi_2})$.

In the supplementary law, since in $\mathbb{F}_p$ we have $c^{(p-1)/2} = (\frac{c}{p}) = \pm 1$ by Euler's congruence in $\mathbb{Z}$ and $(p^d-1)/(p-1) = 1+p+\cdots p^{d-1} \equiv d \bmod 2$,

$$c^{(\mathrm{N}(\pi)-1)/2} = c^{\frac{p-1}{2}\frac{\mathrm{N}(\pi)-1}{p-1}} = \left(\frac{c}{p}\right)^{(\mathrm{N}(\pi)-1)/(p-1)} = \left(\frac{c}{p}\right)^{\deg \pi}.$$

The Legendre symbol in $\mathbb{F}_p[T]$ generalizes to a Jacobi symbol defined similarly to its analogue in $\mathbb{Z}$:

Definition 3.10. Let g be nonzero in $\mathbb{F}_p[T]$. Write $g = c\pi_1\pi_2\ldots\pi_k$, where c is a nonzero constant and the π_i's are monic irreducibles. For f in $\mathbb{F}_p[T]$, define the *Jacobi symbol*

$$\left(\frac{f}{g}\right) = \left(\frac{f}{\pi_1}\right)\left(\frac{f}{\pi_2}\right)\cdots\left(\frac{f}{\pi_k}\right),$$

where each $(\frac{f}{\pi_i})$ is a Legendre symbol. In particular, if g is constant, then $(\frac{f}{g}) = 1$ for nonzero f and $(\frac{0}{g}) = 0$.

When f and g are relatively prime, each $(\frac{f}{\pi_i})$ in Definition 3.10 is ± 1, so $(\frac{f}{g}) = \pm 1$. When f and g are not relatively prime, some monic irreducible dividing g also divides f, which makes some $(\frac{f}{\pi_i})$ equal to 0, so $(\frac{f}{g}) = 0$. Thus $(\frac{f}{g}) = 0$ if and only if f and g are not relatively prime.

Calculating Jacobi symbols in $\mathbb{F}_p[T]$ is as computationally cheap as in $\mathbb{Z}$ because of the following main law and supplementary law of Jacobi reciprocity in $\mathbb{F}_p[T]$.

Theorem 3.11. *For monic $f, g \in \mathbb{F}_p[T]$,*

$$\left(\frac{g}{f}\right) = (-1)^{\frac{\mathrm{N}(f)-1}{2}\cdot\frac{\mathrm{N}(g)-1}{2}} \left(\frac{f}{g}\right).$$

For nonzero $c \in \mathbb{F}_p$ and $g \in \mathbb{F}_p[T]$,

$$\left(\frac{c}{g}\right) = c^{(\mathrm{N}(g)-1)/2}.$$

Proof. If $f = 1$ then both sides of the first equation are 1 for all monic g and similarly if $g = 1$ and f is any monic polynomial. Thus we can now assume f and g are both nonconstant.

When f and g are monic and nonconstant, we can write them each as products of monic irreducibles: $f = \pi_1\pi_2\dots\pi_m$ and $g = \tau_1\tau_2\dots\tau_n$. Then

$$\begin{aligned}\left(\frac{g}{f}\right) &= \prod_{1\le i\le m}\left(\frac{g}{\pi_i}\right)\\ &= \prod_{1\le i\le m}\prod_{1\le j\le n}\left(\frac{\tau_j}{\pi_i}\right)\\ &= \prod_{1\le j\le n}\prod_{1\le i\le m}\left(\frac{\tau_j}{\pi_i}\right).\end{aligned}$$

By quadratic reciprocity for the Legendre symbol in $\mathbb{F}_p[T]$ and Remark 3.9,

$$\begin{aligned}\left(\frac{g}{f}\right) &= \prod_{1\le j\le n}\prod_{1\le i\le m}(-1)^{\frac{p-1}{2}(\deg\pi_i)(\deg\tau_j)}\left(\frac{\pi_i}{\tau_j}\right)\\ &= (-1)^S\prod_{1\le j\le n}\left(\frac{f}{\tau_j}\right)\\ &= (-1)^S\left(\frac{f}{g}\right),\end{aligned}$$

where

$$\begin{aligned}S &= \frac{p-1}{2}\sum_{1\le i\le m,1\le j\le n}\deg(\pi_i)\deg(\tau_j)\\ &= \frac{p-1}{2}\left(\sum_{i=1}^{m}\deg(\pi_i)\right)\left(\sum_{j=1}^{n}\deg(\tau_j)\right)\\ &= \left(\frac{p-1}{2}\right)\deg(f)\deg(g)\\ &\equiv \frac{\mathrm{N}(f)-1}{2}\cdot\frac{\mathrm{N}(g)-1}{2} \bmod 2 \quad \text{by Remark 3.9.}\end{aligned}$$

Now, we prove the supplementary law.

Let $c \in \mathbb{F}_p^\times$. We can assume g is monic since the leading coefficient does not affect the value of the Jacobi symbol. First, we consider the case when g is a nonzero constant, so we can use $g = 1$. Then by the definition of the Jacobi symbol in this case, $(\frac{c}{g}) = 1$. Also $\mathrm{N}(g) = 1$, so $c^{(\mathrm{N}(g)-1)/2} = c^0 = 1$.

Now let g be nonconstant. Since we can assume g is monic, write $g = \pi_1\pi_2\dots\pi_k$. Then

$$\left(\frac{c}{g}\right) = \left(\frac{c}{\pi_1}\right)\left(\frac{c}{\pi_2}\right)\cdots\left(\frac{c}{\pi_k}\right).$$

For each $1 \le i \le k$, $(\frac{c}{\pi_i}) \equiv c^{(\mathrm{N}(\pi_i)-1)/2} \bmod \pi_i$. Both sides of this congruence are constants, and two elements of $\mathbb{F}_p$ that are congruent mod g must be equal since g is nonconstant, so $(\frac{c}{\pi_i}) = c^{(\mathrm{N}(\pi_i)-1)/2}$ in $\mathbb{F}_p$ for all i. Thus,

$$\left(\frac{c}{g}\right) = \prod_i c^{(\mathrm{N}(\pi_i)-1)/2}$$

in $\mathbb{F}_p$. Since p is odd and $c^{(p-1)/2} = (\frac{c}{p}) = \pm 1$ in $\mathbb{F}_p$,

$$\begin{aligned} c^{(\mathrm{N}(\pi_i)-1)/2} &= c^{(p^{\deg \pi_i}-1)/2} \\ &= c^{\left(\frac{p-1}{2}\right)(1+p+\cdots+p^{\deg \pi_i - 1})} \\ &= \left(\frac{c}{p}\right)^{(1+p+\cdots+p^{\deg \pi_i - 1})} \\ &= \left(\frac{c}{p}\right)^{\deg \pi_i}. \end{aligned}$$

Thus,

$$\left(\frac{c}{g}\right) = \prod_i c^{(\mathrm{N}(\pi_i)-1)/2} = \prod_i \left(\frac{c}{p}\right)^{\deg \pi_i} = \left(\frac{c}{p}\right)^{\deg g}$$

since $\deg g = \sum_i \deg \pi_i$. □

This lays the groundwork for the Solovay–Strassen primality test in $\mathbb{F}_p[T]$.

3.3 Solovay–Strassen in $\mathbb{F}_p[T]$

From the previous section, we have the following result.

Theorem 3.12. *For an odd prime p and nonconstant $g \in \mathbb{F}_p[T]$, if there exists an $a \in \mathbb{F}_p[T]$ such that $a \not\equiv 0 \bmod g$ and*

$$a^{(\mathrm{N}(g)-1)/2} \not\equiv \left(\frac{a}{g}\right) \bmod g,$$

then g is reducible.

Definition 3.13. For nonconstant $g \in \mathbb{F}_p[T]$, an *Euler witness* for g is an a in $\mathbb{F}_p[T]$ with $\deg a < \deg g$ such that either $(\frac{a}{g}) = 0$ or $a^{(\mathrm{N}(g)-1)/2} \not\equiv (\frac{a}{g}) \bmod g$ if $\gcd(a, g) = 1$. The other a in $\mathbb{F}_p[T]$ with $\deg a < \deg g$ all satisfy $a^{(\mathrm{N}(g)-1)/2} \equiv (\frac{a}{g}) \bmod g$ and $\gcd(a, g) = 1$, and these are called *Euler nonwitnesses* for g.

An unfortunate defect of the Fermat test is that sometimes there are composite numbers or reducible polynomials with no Fermat witnesses that are relatively prime to the modulus (these are the Carmichael numbers or Carmichael polynomials), and in these cases there can be proportionally few witnesses. However, the Solovay–Strassen test has no such issue. This was shown by Solovay and Strassen in $\mathbb{Z}$, and by adapting their method we will show a similar result in $\mathbb{F}_p[T]$ when p is an odd prime.

Theorem 3.14. *For an odd prime p, each reducible $g \in \mathbb{F}_p[T]$ has an Euler witness a that is also relatively prime to g.*

Proof. We want to show that there is an $a \in \mathbb{F}_p[T]$ with $\deg a < \deg g$ such that $\gcd(a, g) = 1$ and $a^{(\mathrm{N}(g)-1)/2} \not\equiv (\frac{a}{g}) \bmod g$.

Case 1: g is not squarefree. Then $g = \pi^e h$ where $e \geq 2$ and $\pi \nmid h$. By the Chinese remainder theorem, there is an a such that $a \equiv 1 + \pi \bmod \pi^e$ and $a \equiv 1 \bmod h$ and $\deg a < \deg g$. Then $\gcd(a, g) = 1$. We will show this a is an Euler witness for g. Since $\gcd(a, g) = 1$, we have $(\frac{a}{g}) = \pm 1$. If $a^{(\mathrm{N}(g)-1)/2} \equiv (\frac{a}{g}) \bmod g$, then square both sides to get $a^{\mathrm{N}(g)-1} \equiv 1 \bmod g$. Since $\pi^2 \mid g$, we can reduce both sides modulo π^2 and get $(1 + \pi)^{\mathrm{N}(g)-1} \equiv 1 \bmod \pi^2$. This leads to the contradiction $\pi \equiv 0 \bmod \pi^2$ by the binomial theorem as in the proof of Theorem 2.19. Thus, $a^{(\mathrm{N}(g)-1)/2} \not\equiv (\frac{a}{g}) \bmod g$, $\gcd(a, g) = 1$, and $\deg a < \deg g$.

Case 2: g is squarefree. Since g has no repeated irreducible factor, $g = c\pi_1\pi_2 \dots \pi_k$, where c is a nonzero constant and the π_i's are distinct monic irreducibles, and $k \geq 2$ since g is reducible. Half of the nonzero elements of $\mathbb{F}_p[T]_{\pi_1}$ are quadratic residues, so there is $b \in \mathbb{F}_p[T]_{\pi_1}$ such that $(\frac{b}{\pi_1}) = -1$. Then, by the Chinese remainder theorem, there is an $a \in \mathbb{F}_p[T]$ such that $a \equiv b \bmod \pi_1$ and $a \equiv 1 \bmod \pi_2 \cdots \pi_k$ and $\deg a < \deg g$. Then $\gcd(a, g) = 1$ and

$$\left(\frac{a}{g}\right) = \left(\frac{a}{\pi_1}\right)\left(\frac{a}{\pi_2}\right) \cdots \left(\frac{a}{\pi_k}\right) = \left(\frac{a}{\pi_1}\right) = -1.$$

If it were true that $a^{(\mathrm{N}(g)-1)/2} \equiv (\frac{a}{g}) \bmod g$, then, reducing modulo π_i for $i \neq 1$ gives us $a^{(\mathrm{N}(g)-1)/2} \equiv -1 \bmod \pi_i$, which is a contradiction since $a \equiv 1 \bmod \pi_i$. Thus, $a^{(\mathrm{N}(g)-1)/2} \not\equiv (\frac{a}{g}) \bmod g$, $\gcd(a, g) = 1$, and $\deg a < \deg g$. □

Since each nonzero congruence class in $\mathbb{F}_p[T]_g$ contains a unique element with degree less than $\deg g$, when saying $a \in \mathbb{F}_p[T]$ is an Euler witness for g we can replace the condition that $\deg a < \deg g$ by saying $a \in \mathbb{F}_p[T]_g$ with $a \neq 0$ in $\mathbb{F}_p[T]_g$. This will be done in the next theorem, which shows for reducible g that the proportion of Euler nonwitnesses for g is less than $1/2$.

Theorem 3.15. *For reducible $g \in \mathbb{F}_p[T]$, the number of its Euler nonwitnesses mod g is less than $\frac{\mathrm{N}(g)-1}{2}$.*

Proof. We show that the subset S of Euler nonwitnesses for g in $\mathbb{F}_p[T]_g$ is a group under multiplication. Saying $a \in \mathbb{F}_p[T]_g$ is an Euler nonwitness for g means $a^{(\mathrm{N}(g)-1)/2} \equiv (\frac{a}{g}) \bmod g$ and $\gcd(a, g) = 1$. If $a, b \in \mathbb{F}_p[T]_g$ are Euler nonwitnesses for g , i.e. $a, b \in S$, then

$$\begin{aligned}(ab)^{(\mathrm{N}(g)-1)/2} &\equiv a^{(\mathrm{N}(g)-1)/2} b^{(\mathrm{N}(g)-1)/2} \bmod g \\ &\equiv \left(\frac{a}{g}\right)\left(\frac{b}{g}\right) \bmod g \\ &\equiv \left(\frac{ab}{g}\right) \bmod g\end{aligned}$$

and $\gcd(ab, g) = 1$ since $\gcd(a, g) = 1$ and $\gcd(b, g) = 1$. So, $ab \in S$ as well. Thus S is closed under multiplication. Furthermore, the identity $1 \in \mathbb{F}_p[T]_g$ is clearly in S. So, it suffices to show that if $a \in S$, then $a^{-1} \in S$, where $aa^{-1} \equiv a^{-1}a \equiv 1 \bmod g$. We know that a^{-1} exists because nonwitnesses are necessarily units, and $\gcd(a^{-1}, g) = 1$. We have

$$\begin{aligned}a^{(\mathrm{N}(g)-1)/2}(a^{-1})^{(\mathrm{N}(g)-1)/2} &\equiv (aa^{-1})^{(\mathrm{N}(g)-1)/2} \bmod g \\ &\equiv 1 \bmod g \\ &\equiv \left(\frac{aa^{-1}}{g}\right) \bmod g \\ &\equiv \left(\frac{a}{g}\right)\left(\frac{a^{-1}}{g}\right) \bmod g.\end{aligned}$$

Since $a^{(\mathrm{N}(g)-1)/2} \equiv (\frac{a}{g}) \bmod g$ (and $(\frac{a}{g}) \neq 0$), this implies $(a^{-1})^{(\mathrm{N}(g)-1)/2} \equiv (\frac{a^{-1}}{g}) \bmod g$. So, S is a group under multiplication. It is a subgroup of the multiplicative group U of the units in $\mathbb{F}_p[T]_g$. Since an Euler witness a for g that is relatively prime to g exists by the previous theorem, $a \in U$ but $a \notin S$, so S is a *proper* subgroup of U. Therefore, by Lagrange's theorem, $|S| \leq \frac{1}{2}|U|$. Since g is reducible, not every nonzero element of $\mathbb{F}_p[T]_g$ is a unit, so $|U| < \mathrm{N}(g) - 1$. Thus, $|S| < \frac{\mathrm{N}(g)-1}{2}$. □

This gives a nice analogue in $\mathbb{F}_p[T]$ of the Solovay–Strassen test in $\mathbb{Z}$. In practice, we can apply the following (even easier) test first to check whether or not a polynomial in $\mathbb{F}_p[T]$ has a repeated irreducible factor.

Theorem 3.16. *A nonconstant $g \in \mathbb{F}_p[T]$ has a repeated irreducible factor if and only if $\gcd(g, g') \neq 1$.*

Proof. Suppose that g has a repeated irreducible factor. Write $g = \pi^2 h$. Then by the product rule,

$$g'(T) = 2\pi(T)\pi'(T)h(T) + \pi(T)^2 h'(T) = \pi(T)(2\pi'(T)h(T) + \pi(T)h'(T)).$$

Since $g(T)$ and $g'(T)$ share the irreducible factor $\pi(T)$, $\gcd(g, g') \neq 1$.

Now suppose g has no repeated irreducible factor. We will show $\gcd(g, g') = 1$, by contradiction. Suppose $\gcd(g, g') \neq 1$. Let π be a common irreducible factor of g and g'. Thus, $g = \pi h$ and $\pi \nmid h$. By the product rule,

$$g'(T) = \pi'(T)h(T) + \pi(T)h'(T).$$

Since $\pi \mid g'$, we get $\pi \mid \pi' h$. Since $\pi \nmid h$, necessarily $\pi \mid \pi'$. But π' is a polynomial that, if it is not 0, has degree less than $\deg \pi$, so it couldn't possibly be divisible by π. Thus the only way we can have $\pi \mid \pi'$ is if $\pi'(T) = 0$. Writing $\pi(T) = \sum_{i=0}^{d} c_i T^i$ with $d \geq 1$, so $\pi'(T) = \sum_{i=1}^{d} i c_i T^{i-1}$, that $\pi'(T) = 0$ means $ic_i = 0$ in $\mathbb{F}_p$ for all $i \geq 1$ and it is also true when $i = 0$. In other words, when $c_i \neq 0$ in $\mathbb{F}_p$ we need $i = 0$ in $\mathbb{F}_p$. That means $p \mid i$ in $\mathbb{Z}$ when $c_i \neq 0$: all the nonzero terms in $\pi(T)$ are in degrees divisible by p: $\pi(T)$ is a polynomial in T^p! Thus $\pi(T) = k(T^p)$ where $k(T) = \sum_{j=0}^{r} a_j T^j$ with $a_j \in \mathbb{F}_p$. By Fermat's little theorem, $a_j = a_j^p$. Thus

$$\pi(T) = k(T^p) = \sum_{j=0}^{r} a_j T^{pj} = \sum_{j=0}^{r} a_j^p T^{pj} = \sum_{j=0}^{r} (a_j T^j)^p. \tag{1}$$

The p-th power is additive in characteristic p, such as in the ring $\mathbb{F}_p[T]$, so (1) implies

$$\pi(T) = \left(\sum_{j=0}^{r} a_j T^j \right)^p$$

in $\mathbb{F}_p[T]$. This contradicts irreducibility of π. Thus when g has no repeated irreducible factor, it is not the case that $\gcd(g, g') \neq 1$, so $\gcd(g, g') = 1$. □

Thus, when testing the irreducibility of g, we can first check if $\gcd(g, g') \neq 1$, since when that happens, g is reducible and we're done. Since taking the gcd by the Euclidean algorithm is cheaper computationally than running the Solovay–Strassen test, so we should only bother using the Solovay–Strassen test, or really any other irreducibility test, on square-free polynomials.

Putting the last two theorems together, here is how the Solovay–Strassen test can be used in practice on $\mathbb{F}_p[T]$. Given a nonconstant $g \in \mathbb{F}_p[T]$, do the following:

1. Check if $\gcd(g, g') \neq 1$ using Euclid's algorithm. If it is, then g is reducible.
2. Choose a random $a \in \mathbb{F}_p[T]$ with $\deg a < \deg g$ that has not been tested so far.
3. Compute $(\frac{a}{g})$ using the main law and supplementary laws of Jacobi reciprocity.
4. If $(\frac{a}{g}) = 0$, then $\gcd(a, g) \neq 1$ and g is reducible.
5. If $(\frac{a}{g}) \neq 0$, then compute $a^{(\mathrm{N}(g)-1)/2} \bmod g$.

6. If $(\frac{a}{g}) \neq 0$ and $a^{(\mathrm{N}(g)-1)/2} \not\equiv (\frac{a}{g}) \bmod g$, then g is reducible. Otherwise, return to step 2.

By Theorem 3.15, if we randomly pick N different residues mod g and always find $a^{(\mathrm{N}(g)-1)/2} \equiv (\frac{a}{g}) \bmod g$ and $(\frac{a}{g}) = \pm 1$, then the likelihood that g is irreducible is at most $1 - (\frac{1}{2})^N$. Even for small values of N like 10 or 20 (increasing the practicality of the computation), the chances that a reducible g has no Euler witness found after N runs of the Solovay–Strassen test become negligible, so we can be nearly certain that g is irreducible.

In some ways, the Solovay–Strassen test is very similar to the Fermat test. It operates by breaking a congruence that holds for irreducibles, but unlike the Fermat test there is no possibility of anything like Carmichael exceptions: the proportion of witnesses for a reducible input is always at least 1/2. The extra cost involved is that we need to compute some Jacobi symbols, but that can be done quickly without factoring thanks to Jacobi reciprocity.

3.4 Solovay–Strassen in Characteristic 2

For p an odd prime, the Solovay–Strassen test in the integers generalizes nicely to $\mathbb{F}_p[T]$ since quadratic reciprocity and then Jacobi reciprocity carry over cleanly from $\mathbb{Z}$ to $\mathbb{F}_p[T]$.

The starting point for quadratic reciprocity in $\mathbb{F}_p[T]$ is the elementary property that exactly half the nonzero elements in $\mathbb{F}_p[T]_\pi$ are quadratic residues when $\pi \in \mathbb{F}_p[T]$ is irreducible. A sketch of the proof for this property is as follows. For $a, b \in \mathbb{F}_p[T]$, the condition $a^2 \equiv b^2 \bmod \pi$ is equivalent to $\pi \mid (a-b)(a+b)$, which is the same as $a \equiv \pm b \bmod \pi$.

However, in $\mathbb{F}_2[T]$, or more generally in $F[T]$ where F is any field of characteristic 2, $a^2 - b^2 = (a-b)^2$, so $\pi \mid a^2 - b^2$ is equivalent to $\pi \mid (a-b)^2$, which is the same as $\pi \mid a - b$ since π is irreducible, and thus $a \equiv b \bmod \pi$. So, squaring on $\mathbb{F}_2[T]_\pi$ is injective, which implies it is surjective since $\mathbb{F}_2[T]_\pi$ is finite. Thus, quadratic reciprocity must be adjusted to be interesting in $\mathbb{F}_2[T]$. We will discuss how this is done and how it changes the mechanism of the Solovay–Strassen test in $\mathbb{F}_2[T]$.

In this section, we will work not just in $\mathbb{F}_2[T]$, but in $\mathbb{F}[T]$ where $\mathbb{F}$ is any finite field with characteristic 2. That makes $|\mathbb{F}| = q$ be a power of 2.

It turns out that, according to resources provided by Keith Conrad for this project, a kind-of-natural substititue for the squaring map is the function $\wp(x) = x^2 + x$ for $x \in \mathbb{F}[T]$. We note that while the square map is multiplicative, i.e. $(xy)^2 = x^2y^2$, this new map $\wp$ is additive:

$$\wp(x+y) = (x+y)^2+(x+y) = (x^2+y^2)+(x+y) = (x^2+x)+(y^2+y) = \wp(x)+\wp(y).$$

We now define characteristic 2 analogues of the Legendre and Jacobi symbols.

Definition 3.17. For finite $\mathbb{F}$ with characteristic 2, irreducible $\pi \in \mathbb{F}[T]$, and any $f \in \mathbb{F}[T]$, the *characteristic 2 quadratic residue symbol* $[f, \pi)$ is defined as

follows:

$$[f,\pi) = \begin{cases} 0, & \text{if } f \equiv x^2 + x \bmod \pi \text{ for some } x \in \mathbb{F}[T], \\ 1, & \text{if } f \not\equiv x^2 + x \bmod \pi \text{ for any } x \in \mathbb{F}[T]. \end{cases}$$

We say f is a *$\wp$-value modulo π* if $[f,\pi) = 0$, and a *non $\wp$-value* if $[f,\pi) = 1$.

We will see in Theorem 3.20 that, unlike the Legendre symbol, $[f,\pi)$ is additive rather than multiplicative in f. This is why we give it values 0 and 1 in characterisic 2. The symbol has a square bracket on the left side to emphasize its asymmetric behavior in f and π in a reciprocity law, which we'll see in Theorem 3.24. The following lemma asserts that, much like with quadratic residues in $\mathbb{Z}$, exactly half the elements of $\mathbb{F}[T]_\pi$ are $\wp$-values. Unlike with $\mathbb{Z}_p$ when p is odd, this count of half in $\mathbb{F}[T]_\pi$ includes 0.

Lemma 3.18. *For irreducible $\pi \in \mathbb{F}[T]$, half the elements (including zero!) of $\mathbb{F}[T]_\pi$ are $\wp$-values:*

$$|\{g^2 + g \bmod \pi \mid g \in \mathbb{F}[T]\}| = \frac{1}{2} q^{\deg \pi}.$$

Proof. We will show the function $\wp : g \bmod \pi \mapsto g^2 + g \bmod \pi$ is 2-to-1 on $\mathbb{F}[T]_\pi$. Since $\mathbb{F}[T]_\pi$ is a field, the only solutions in it of $x^2 = x$ are 0 and 1. So for g and h in $\mathbb{F}[T]$,

$$\begin{aligned} \wp(g) \equiv \wp(h) \bmod \pi &\iff g^2 + g \equiv h^2 + h \bmod \pi \\ &\iff (g-h)^2 \equiv g - h \bmod \pi \\ &\iff g - h \equiv 0 \text{ or } 1 \bmod \pi \\ &\iff g \equiv h \text{ or } h + 1 \bmod \pi. \end{aligned}$$

Thus, in $\mathbb{F}[T]_\pi$, each $\wp$-value arises in exactly two ways from $\mathbb{F}[T]_\pi$: when it is $\wp(g)$, it is also $\wp(g+1)$; note $\wp((g+1)+1) = \wp(g)$ brings us back to $\wp(g)$. Thus, just as the equivalence $a^2 \equiv b^2 \bmod p \iff a \equiv \pm b \bmod p$ implies the number of nonzero squares in $\mathbb{Z}_p^\times$ is $|\mathbb{Z}_p^\times|/2 = (p-1)/2$, the image of $\wp$ on $\mathbb{F}[T]_\pi$ has half the size of $\mathbb{F}[T]_\pi$. Since $|\mathbb{F}[T]_\pi| = q^{\deg \pi}$, the number of $\wp$-values in $\mathbb{F}[T]_\pi$ is $q^{\deg \pi}/2$. □

Corollary 3.19. *For irreducible $\pi \in \mathbb{F}[T]$, half of the f in $\mathbb{F}[T]_\pi$ satisfy $[f,\pi) = 0$ and half satisfy $[f,\pi) = 1$.*

The characteristic 2 residue symbol admits some interesting properties, similar to the properties of the Legendre symbol in $\mathbb{Z}_p$.

Theorem 3.20. *For irreducible $\pi \in \mathbb{F}[T]$, the symbol $[f,\pi)$ has the following properties:*

1. *if $f_1 \equiv f_2 \bmod \pi$, then $[f_1,\pi) = [f_2,\pi)$,*
2. *$f + f^2 + f^4 + f^8 + \cdots + f^{q^{\deg \pi}/2} \equiv [f,\pi) \bmod \pi$,*

3. $[f_1 + f_2, \pi) = [f_1, \pi) + [f_2, \pi)$.

Proof. See [5] for proofs. □

The second property serves as the analogue of Euler's congruence in $\mathbb{Z}$ and $\mathbb{F}_p[T]$ for odd primes p.

We can extend the new symbol $[f, \pi)$ to a new Jacobi symbol $[f, g)$ as well.

Definition 3.21. For $f, g \in \mathbb{F}[T]$ with $g \neq 0$, write $g = c\pi_1\pi_2 \dots \pi_k$ for $c \in \mathbb{F}^\times$ and monic irreducibles π_i. Define the characteristic 2 Jacobi symbol

$$[f, g) = [f, \pi_1) + [f, \pi_2) + \cdots + [f, \pi_k)$$

in $\mathbb{F}_2$. In particular, $[f, c) = 0$ for $c \in \mathbb{F}^\times$.

The symbol $[f, g)$ satisfies properties similar to the symbol $[f, \pi)$, excluding anything like the second property in Theorem 3.20 (which is what will lead to the Solovay–Strassen test in $\mathbb{F}[T]$) and adding a new property that turns multiplication in the second component into addition in $\mathbb{F}_2$. These properties all follow from the definitions and properties of the characteristic 2 residue symbol:

1. if $f_1 \equiv f_2 \bmod g$, then $[f_1, g) = [f_2, g)$,
2. $[f_1 + f_2, g) = [f_1, g) + [f_2, g)$,
3. $[f, g_1 g_2) = [f, g_1) + [f, g_2)$.

The symbol $[f, g)$ has an analogue of quadratic reciprocity, but it is expressed very differently; we cannot simply "flip" the components in the symbol, but rather reduce the symbol in two different ways.

Definition 3.22. For $g(T) \in \mathbb{F}[T]$, define

$$g^*(T) = \frac{g(T)}{T^{\deg g}}.$$

Example 3.23. If $g(T) = T^4 + T^2 + T + 1$ in $\mathbb{F}_2[T]$, then $g^* = \frac{1}{T^4} + \frac{1}{T^3} + \frac{1}{T^2} + 1$.

We have the following theorem (see [5] for the proof and background theory).

Theorem 3.24. *If $q = |\mathbb{F}| = 2^m$, then for nonzero $g \in \mathbb{F}[T]$, the following supplementary law and main law hold: for $c \in \mathbb{F}$,*

$$[c, g) \equiv (\deg g)(c + c^2 + c^4 + \cdots + c^{2^{m-1}}) \bmod 2,$$

and for nonconstant $f \in \mathbb{F}[T]$ with $f(0) = 0$,

$$g_1^* \equiv g_2^* \bmod 1/T^{\deg f + 1} \implies [f, g_1) = [f, g_2).$$

This allows us to reduce a Jacobi symbol $[f, g)$ in two ways when $f \neq 0$ and g is nonconstant: if $\deg f > \deg g$, then we reduce f modulo g. If $\deg f < \deg g$, then write $f(T) = c + h(T)$ where $c = f(0)$ and $h(0) = 0$, so $[f, g) = [c, g) + [h, g)$. We can compute $[c, g)$ by the supplementary law. If $h(T) \neq 0$, then $\deg h = \deg f < \deg g$ and we reduce g^* modulo $1/T^{\deg h+1}$, which is $1/T^{\deg f+1}$, to rewrite $[h, g)$ as $[h, \widetilde{g})$ with a new $\widetilde{g}$ where either $\deg \widetilde{g} < \deg g$ or $\widetilde{g} = 0$.

Example 3.25. We compute $[f, g)$ in $\mathbb{F}_2[T]$, where $f = T^3+1$ and $g = T^5+T^4+T^2+T+1$. When $c \in \mathbb{F}_2$, the supplementary law says $[c, h) \equiv (\deg h)c \bmod 2$.

By additivity and the supplementary law,

$$[f, g) = [T^3, g) + [1, g) = [T^3, g) + 1. \tag{2}$$

Since $\deg f < \deg g = 5$, we consider $g^* = g/T^5 = \frac{1}{T^5} + \frac{1}{T^4} + \frac{1}{T^3} + \frac{1}{T} + 1$ modulo $\frac{1}{T^4}$: $g^* \equiv \frac{1}{T^3} + \frac{1}{T} + 1 \bmod \frac{1}{T^4}$ and $\frac{1}{T^3} + \frac{1}{T} + 1 = (T^3 + T^2 + 1)^*$, so by the main law in Theorem 3.24 we have

$$[T^3, g) = [T^3, T^3 + T^2 + 1). \tag{3}$$

Since $T^3 \equiv T^2 + 1 \bmod T^3 + T^2 + 1$, in the first component in (3) we can replace T^3 with $T^2 + 1$, so by (2) and (3)

$$[f, g) = [T^2 + 1, T^3 + T^2 + 1) + 1. \tag{4}$$

By additivity and the supplementary law,

$$[T^2+1, T^3+T^2+1) = [T^2, T^3+T^2+1) + [1, T^3+T^2+1) = [T^2, T^3+T^2+1) + 1,$$

so by (4),

$$[f, g) = [T^2, T^3 + T^2 + 1) + 1 + 1 = [T^2, T^3 + T^2 + 1). \tag{5}$$

Since $(T^3 + T^2 + 1)^* = \frac{1}{T^3} + \frac{1}{T} + 1 \equiv \frac{1}{T} + 1 \bmod \frac{1}{T^3}$ and $\frac{1}{T} + 1 = (T + 1)^*$, by the main law we have $[T^2, T^3 + T^2 + 1) = [T^2, T + 1)$.

Since $T^2 \equiv 1 \bmod T + 1$, $[T^2, T + 1) = [1, T + 1)$, which is 1 by the supplementary law. Thus by (5)

$$[f, g) = [T^2, T^3 + T^2 + 1) = [T^2, T + 1) = [1, T + 1) = 1,$$

which completes the calculation.

The formulas in Theorem 3.24 allow us to compute Jacobi symbols in characteristic 2 in a double-Euclidean-algorithm method. In this process, rather than taking advantage of a flipping feature in the residue symbol, we exploit a periodicity in the second component.

Such periodicity happens with the Jacobi symbol in $\mathbb{Z}$ too. For a nonzero integer a, the value of $(\frac{a}{n})$ (for positive odd n) is determined by n modulo $4|a|$ (this follows from Jacobi reciprocity). So, if m and n are positive odd numbers and $m \equiv n \bmod 4|a|$, then $(\frac{a}{m}) = (\frac{a}{n})$. In characteristic 2 we met a similar property, except it works modulo a polynomial in $1/T$, which is still very strange.

In $\mathbb{F}[T]$, we define Euler witnesses for g as polynomials in $\mathbb{F}[T]_g$ that break the analogue of Euler's congruence, as follows.

Definition 3.26. For nonconstant $g \in \mathbb{F}[T]$, an *Euler witness* for g is an $a \in \mathbb{F}[T]_g$ such that

$$a + a^2 + \cdots + a^{q^{\deg g}/2} \not\equiv [a, g) \bmod g.$$

An $a \in \mathbb{F}[T]_g$ such that $a + a^2 + \cdots + a^{q^{\deg g}/2} \equiv [a, g) \bmod g$ is an *Euler nonwitness* for g.

Irreducible g in $\mathbb{F}[T]$ have no Euler witness. Now, we state and prove for reducible g in $\mathbb{F}[T]$ analogues of what led to a Solovay–Strassen test in $\mathbb{F}_p[T]$ when p is odd.

Theorem 3.27. *Let $g \in \mathbb{F}[T]$ be reducible. There exists an Euler witness for g.*

Proof. Note that $\mathbb{F}[T]$ has unique factorization. Moreover, every element in $\mathbb{F}$ is a square, so we can write $g = GH^2$, where G is squarefree and monic. Then for all a in $\mathbb{F}[T]$, $[a, g) = [a, GH^2) = [a, G) + 2[a, H) = [a, G)$.

Case 1: $G = 1$. Let $a = H$, which is nonconstant since g is reducible. Then $[a, g) = [a, G) = [a, 1) = 0$, while

$$a + a^2 + \cdots + a^{q^{\deg g}/2} = H + H^2 + \cdots \equiv H \bmod H^2.$$

If $a + a^2 + \cdots + a^{q^{\deg g}/2} \equiv [a, g) \bmod g$, then we can reduce both sides modulo H^2 since $H^2 \mid g$ and we get $H \equiv 0 \bmod H^2$, which is a contradiction since H is nonconstant. Thus H is an Euler witness for g.

Case 2: $G = \pi$ for a monic irreducible π that divides H, so $\pi^2 \mid g$. Let $a = \pi$. Then $[a, g) = [a, G) = [\pi, \pi) = 0$, while

$$a + a^2 + \cdots + a^{q^{\deg g}/2} = \pi + \pi^2 + \cdots \equiv \pi \bmod \pi^2.$$

If $a + a^2 + \cdots + a^{q^{\deg g}/2} \equiv [a, g) \bmod g$, then reducing both sides modulo π^2 implies $\pi \equiv 0 \bmod \pi^2$. This is a contradiction, so π is an Euler witness for g.

Case 3: $G = \pi$ for a monic irreducible π that doesn't divide H. Since g is reducible, H is nonconstant. Let τ be an irreducible factor of H, so π and τ are relatively prime. Let $b \in \mathbb{F}[T]_\pi$ be a non $\wp$-value, so $[b, g) = [b, G) = [b, \pi) = 1$. By the Chinese remainder theorem there is an $a \in \mathbb{F}[T]$ such that $a \equiv b \bmod \pi$ and $a \equiv 0 \bmod \tau$. Then $[a, g) = [a, G) = [a, \pi) = [b, \pi) = 1$, while

$$a + a^2 + \cdots + a^{q^{\deg g}/2} \equiv 0 \bmod \tau.$$

If $a + a^2 + \cdots + a^{q^{\deg g}/2} \equiv [a, g) \bmod g$, then reducing both sides modulo τ implies $0 \equiv 1 \bmod \tau$, which is a contradiction. Thus a is an Euler witness for g.

Case 4: $G = \pi_1 \pi_2 \dots \pi_k$ for distinct monic irreducibles π_i such that $k \geq 2$. Let $b \in \mathbb{F}[T]_{\pi_1}$ be a non $\wp$-value, so $[b, \pi_1) = 1$. By the Chinese remainder theorem there is an $a \in \mathbb{F}[T]$ such that $a \equiv b \bmod \pi_1$ and $a \equiv 0 \bmod \pi_i$ for each $i \geq 2$. Then $[a, g) = [a, G) = \sum_{i=1}^{k} [a, \pi_i) = [a, \pi_1) = [b, \pi_1) = 1$, while

$$a + a^2 + \cdots + a^{q^{\deg g}/2} \equiv 0 \bmod \pi_2.$$

If $a + a^2 + \cdots + a^{q^{\deg g}/2} \equiv [a, g) \bmod g$, then reducing both sides modulo π_2 implies $0 \equiv 1 \bmod \pi_2$. This is a contradiction, so a is an Euler witness for g. $\square$

Remark 3.28. Unlike in Theorem 3.14, we don't require that a be relatively prime to g in Theorem 3.27. This is because, as will be shown in the next theorem, the subset of Euler nonwitnesses for g in $\mathbb{F}[T]_g$ is an additive subgroup because of the additive structure of the Jacobi symbol in characteristic 2. Therefore, there is no reason to require that Euler witnesses be relatively prime to the modulus in order to show that having one Euler witness leads to having many Euler witnesses.

Theorem 3.29. *Let $g \in \mathbb{F}[T]$ be reducible. The number of Euler nonwitnesses for g in $\mathbb{F}[T]_g$ is at most $\frac{\mathrm{N}(g)}{2}$.*

Proof. Let

$$S = \{a \in \mathbb{F}[T]_g \mid a + a^2 + \cdots + a^{q^{\deg g}/2} \equiv [a, g) \bmod g\}.$$

We show that S is a group under *addition*. Clearly, $0 \in S$. Suppose $a, b \in S$. Then $a+a^2+\cdots+a^{q^{\deg g}/2} \equiv [a, g) \bmod g$ and $b+b^2+\cdots+b^{q^{\deg g}/2} \equiv [b, g) \bmod g$. Since we are working in characteristic 2, $(a + b)^2 = a^2 + b^2$, and, inductively, $(a + b)^{2^n} = a^{2^n} + b^{2^n}$ for all $n \geq 1$. So,

$$\begin{aligned}
&(a + b) + (a + b)^2 + \cdots + (a + b)^{q^{\deg g}/2} \\
&= (a + b) + (a^2 + b^2) + \cdots + (a^{q^{\deg g}/2} + b^{q^{\deg g}/2}) \\
&= (a + a^2 + \cdots + a^{q^{\deg g}/2}) + (b + b^2 + \cdots + b^{q^{\deg g}/2}) \\
&\equiv [a, g) + [b, g) \bmod g \\
&\equiv [a + b, g) \bmod g,
\end{aligned}$$

and thus $a + b \in S$ as well. Clearly, if $a \in S$, then $-a = a \in S$, so therefore, S is an additive subgroup of $\mathbb{F}[T]_g$. By the previous theorem, there exists an Euler witness for g in $\mathbb{F}[T]_g$, so S is a proper subgroup of $\mathbb{F}[T]_g$. By Lagrange's theorem, $|S| \leq \frac{1}{2}|\mathbb{F}[T]_g| = \frac{\mathrm{N}(g)}{2}$. □

4 The Miller–Rabin Test

4.1 The Miller Rabin Test in $\mathbb{Z}$

Let $n > 1$ be an odd number. Then factor out the largest power of 2 from $n-1$, so $n - 1 = 2^e k$ where $e \geq 1$, and k is odd.
The intution behind the Miller–Rabin test can be shown by repeatedly factoring

the polynomial $x^{n-1} - 1 = x^{2^e k} - 1$:

$$\begin{aligned} x^{2^e k} - 1 &= (x^{2^{e-1} k})^2 - 1 \\ &= (x^{2^{e-1} k} - 1)(x^{2^{e-1} k} + 1) \\ &= (x^{2^{e-2} k} - 1)(x^{2^{e-2} k} + 1)(x^{2^{e-1} k} + 1) \\ &\vdots \\ &= (x^k - 1)(x^k + 1)(x^{2k} + 1)(x^{4k} + 1) \dots (x^{2^{e-1} k} + 1) \end{aligned}$$

If n is prime and $1 \leq a \leq n-1$, then by Fermat's little theorem $a^{n-1} - 1 \equiv 0 \bmod n$. So we have $(a^k - 1)(a^k + 1)(a^{2k} + 1)(a^{4k} + 1) \dots (a^{2^{e-1} k} + 1) \equiv 0 \bmod n$. Thus, if n is prime, one of these factors must be $0 \bmod n$, so

$$a^k \equiv 1 \bmod n \text{ or } a^{2^i k} \equiv -1 \bmod n \text{ for some } i \in \{0, \dots, e-1\}.$$

These congruences depend on a only modulo n and none hold when $a = 0$, so to check them we can focus on a in the range $\{1, 2, \dots, n-1\}$.

Definition 4.1. Let n be an odd integer greater than 1 and write $n - 1 = 2^e k$ for $e \geq 1$ and an odd integer $k \geq 1$. An integer a in $\{1, 2, \dots, n-1\}$ is called a *Miller–Rabin witness* for n if $a^k \not\equiv 1 \bmod n$ and $a^{2^i k} \not\equiv -1 \bmod n$ for all $i \in \{0, \dots, e-1\}$. We say that a a *Miller–Rabin nonwitness* for n if either $a^k \equiv 1 \bmod n$ or $a^{2^i k} \equiv -1 \bmod n$ for some $i \in \{0, \dots, e-1\}$.

We call $(a^k, a^{2k}, a^{4k}, \dots, a^{2^{e-1} k}) \bmod n$ the *Miller–Rabin* sequence for n that is generated by a.

Odd primes have no Miller–Rabin witness, so if n has at least one Miller–Rabin witness, then n is composite.

Example 4.2. Let $n = 9$. Then, $n - 1 = 8$. So $e = 3$, $k = 1$, and the Miller–Rabin sequence generated by a is $(a, a^2, a^4) \bmod 9$.

$a \bmod 9$	1	2	3	4	5	6	7	8
$a^2 \bmod 9$	1	4	0	7	7	0	4	1
$a^4 \bmod 9$	1	7	0	4	4	0	7	1

As shown by the table above, $2, 3, 4, 5, 6$, and 7 are Miller–Rabin witnesses for 9. Thus, 9 is composite with numerous Miller–Rabin witnesses.

Rabin [6] showed that for any odd composite $n > 1$, the proportion of $a \in \{1, \dots, n-1\}$ that are Miller–Rabin witnesses for n is at least 75%, or equivalently the proportion of Miller–Rabin nonwitnesses for n is at most 25%.

4.2 The Miller–Rabin Test in $\mathbb{F}_p[T]$

The Miller–Rabin test can be adapted to $\mathbb{F}_p[T]$ if p is an odd prime. Let $f \in \mathbb{F}_p[T]$ be monic and nonconstant. Since $\mathrm{N}(f) = p^{\deg f}$ is odd and greater than 1, we can write $\mathrm{N}(f) - 1 = 2^e k$ where $e \geq 1$ and k is odd.

Definition 4.3. When $f \in \mathbb{F}_p[T]$ is monic and nonconstant, let $\mathrm{N}(f) - 1 = 2^e k$ as above. A nonzero $a \in \mathbb{F}_p[T]$ with $\deg a < \deg f$ is called a *Miller–Rabin witness* for f if $a^k \not\equiv 1 \bmod f$ and $a^{2^i k} \not\equiv -1 \bmod f$ for all $i \in \{0, 1, \dots e-1\}$.

Similarly, $a \in \mathbb{F}_p[T]$ with $\deg a < \deg f$ is called a *Miller–Rabin nonwitness* for f if either $a^k \equiv 1 \bmod f$ or $a^{2^i k} \equiv -1 \bmod f$ for some $i \in \{0, 1, \dots e-1\}$.

An irreducible f has no Miller–Rabin witness by Theorem 2.12 in the same way that we saw an odd prime has no Miller–Rabin witness by Fermat's little theorem. Thus if a monic nonconstant f has a Miller–Rabin witness, then f *must* be reducible.

Example 4.4. Consider $f(T) = T^2 - 1$ in $\mathbb{F}_5[T]$. Then, $\mathrm{N}(f) = 5^2 - 1 = 24 = 2^3 \times 3$. So, $e = 3$ and $k = 3$
Now, we want to consider the Miller–Rabin sequence $(a^k, a^{2k}, \dots, a^{2^{e-1}k}) = (a^3, a^6, a^{12}) \bmod f$.
Note that $T^2 \equiv 1 \bmod f$, so we can reduce T^2 to 1 any time it appears in our calculations:

$a(T)$	$a^3 \bmod f$	$a^6 \bmod f$	$a^{12} \bmod f$
$T+1$	$4T+4$	$2T+2$	$3T+3$
$4T+1$	$T+4$	$3T+2$	$2T+3$
$2T+2$	$2T+2$	$3T+3$	$3T+3$
$3T+2$	$3T+2$	$2T+3$	$2T+3$.

As shown in the table above, for all $a \in \{T+1, 4T+1, 2T+2, 3T+2\}$ and all $i \in \{3, 6, 12\}$, $a^i \not\equiv \pm 1 \bmod f$. So $T+1, 4T+1, 2T+2$, and $3T+2$ are all Miller–Rabin witnesses for f. Hence, f is reducible.

Every a with $\deg a < \deg f$ that is a Miller–Rabin nonwitness for f satisfies $a^{\mathrm{N}(f)-1} \equiv 1 \bmod f$, so a is a Fermat nonwitness for f. Thus, by using the contrapositive, every Fermat witness for f is a Miller–Rabin witness for f.

While not strictly necessary for proving a lower bound on the proportion of Miller–Rabin witnesses for odd composite numbers, the following theorem provides a constructive method of finding a Miller–Rabin witness in $\mathbb{F}_p[T]$.

Theorem 4.5. *If a monic nonconstant f is reducible in $\mathbb{F}_p[T]$, then there exists a Miller–Rabin witness a in $\mathbb{F}_p[T]$ for f such that* $\gcd(a, f) = 1$.

Proof. We divide the proof into two cases: when f is Carmichael and when it is not.

Case 1: f is not Carmichael.

There necessarily exists an $a \in \mathbb{F}_p[T]_f^\times$ such that $a^{\mathrm{N}(f)-1} \not\equiv 1 \bmod f$. This means a is a Fermat witness for f, so a is a Miller–Rabin witness for f, and $\gcd(a, f) = 1$.

Case 2: f is Carmichael.

By Korselt's criterion for polynomials (see Section 2.4), f is squarefree, so we can write $f = \pi_1 \dots \pi_k$ for distinct monic irreducibles π_i with $k \geq 2$. Write

$\mathrm{N}(\pi_1)-1 = 2^v k_1$. Choose a generator g for $\mathbb{F}_p[T]^\times_{\pi_1}$ (for existence, see [2] and [3]). By the Chinese remainder theorem, there exists $a \in \mathbb{F}_p[T]$ with $\deg a < \deg f$ such that

$$a \equiv g^{k_1} \bmod \pi_1 \text{ and } a \equiv 1 \bmod \pi_i$$

for all $i \neq 1$. Then $\gcd(a, f) = 1$.

To show a is a Miller–Rabin witness for f, assume it is not. Write $\mathrm{N}(f)-1 = 2^e k$. Then $a^k \equiv 1 \bmod f$ or $a^{2^i k} \equiv -1 \bmod f$ for some $i \in \{0, 1, \dots e-1\}$. In the first case, $a^k \equiv 1 \bmod \pi_1$, so $g^{kk_1} \equiv 1 \bmod \pi_1$, so $\mathrm{N}(\pi_1) - 1 \mid kk_1$, which is a contradiction since $\mathrm{N}(\pi_1) - 1$ is even and kk_1 is odd. In the second case, $a^{2^i k} \equiv -1 \bmod \pi_2$ for some $i \leq e-1$, so $1 \equiv -1 \bmod \pi_2$, which is a contradiction. □

Lemma 4.6. *If a, b in $\mathbb{F}_p[T]$ satisfy $a \equiv b \bmod \pi$, then $a^{\mathrm{N}(\pi)} \equiv b^{\mathrm{N}(\pi)} \bmod \pi^2$.*

Proof. This is true when $a \equiv b \equiv 0 \bmod \pi$ since $\mathrm{N}(\pi) \geq 2$, so we now may assume $a \equiv b \not\equiv 0 \bmod \pi$.

We have

$$\begin{aligned} a^{\mathrm{N}(\pi)} - b^{\mathrm{N}(\pi)} &= a^{p^{\deg \pi}} - b^{p^{\deg \pi}} \\ &= (a-b)(a^{p^{\deg \pi}-1} + a^{p^{\deg \pi}-2}b + \cdots + b^{p^{\deg \pi}-1}). \end{aligned} \tag{6}$$

In the second factor in (6), the exponents in each term sum to $p^{\deg \pi} - 1$. Since $a \equiv b \bmod \pi$, $a^{(p^{\deg \pi}-1-k)}b^k \equiv a^{p^{\deg \pi}-1} \equiv 1 \bmod \pi$ for all $k \in \{0, \dots, p^{\deg \pi}-1\}$, so

$$\begin{aligned} a^{p^{\deg \pi}-1} + a^{p^{\deg \pi}-2}b + \cdots + b^{p^{\deg \pi}-1} &\equiv 1 + 1 + \cdots + 1 \bmod \pi \quad (p^{\deg \pi} \text{ ones}) \\ &\equiv p^{\deg \pi} \bmod \pi \\ &\equiv 0 \bmod \pi \end{aligned}$$

since $p = 0$ in $\mathbb{F}_p[T]$. Thus both factors in (6) are divisible by π, so

$$a^{\mathrm{N}(\pi)} - b^{\mathrm{N}(\pi)} \equiv 0 \bmod \pi^2,$$

which gives us $a^{\mathrm{N}(\pi)} \equiv b^{\mathrm{N}(\pi)} \bmod \pi^2$. □

Lemma 4.7. *Let $f \in \mathbb{F}_p[T]$ be nonzero with a repeated irreducible factor π. If $a^{\mathrm{N}(f)-1} \equiv 1 \bmod f$ and $a \equiv b \bmod \pi$, then $a \equiv b^{\mathrm{N}(f)} \bmod \pi^2$.*

Proof. Since $a^{\mathrm{N}(f)-1} \equiv 1 \bmod f$, $a^{\mathrm{N}(f)} \equiv a \bmod f$. Then $a^{\mathrm{N}(f)} \equiv a \bmod \pi^2$ because $\pi^2 \mid f$. From Lemma 4.6, we have $a^{\mathrm{N}(\pi)} \equiv b^{\mathrm{N}(\pi)} \bmod \pi^2$. Raising both sides to the $\mathrm{N}(f)/\mathrm{N}(\pi)$-power,

$$a^{\mathrm{N}(f)} \equiv b^{\mathrm{N}(f)} \bmod \pi^2.$$

Thus

$$a \equiv a^{\mathrm{N}(f)} \equiv b^{\mathrm{N}(f)} \bmod \pi^2.$$ □

Lemma 4.8. *For monic irreducible polynomial $\pi \in \mathbb{F}_p[T]$, an odd integer k, and $x \in \mathbb{F}_p[T]$, we have*

$$x^k \equiv -1 \bmod \pi \iff x^{(k,\mathrm{N}(\pi)-1)} \equiv -1 \bmod \pi.$$

Proof. ($\Rightarrow$) Bezout's theorem guarantees that we can write $km + (\mathrm{N}(\pi)-1)n = (k, \mathrm{N}(\pi)-1)$ for some integers m and n. Reduce the equation modulo 2 to get that $km \equiv 1 \bmod 2$, hence m is odd. Now, we have

$$x^{(k,\mathrm{N}(\pi)-1)} \equiv x^{km+(\mathrm{N}(\pi)-1)n} \equiv x^{km} \equiv (-1)^m \equiv -1 \bmod \pi.$$

($\Leftarrow$) Observe that k is an odd multiple of $(k, \mathrm{N}(\pi)-1)$. □

Theorem 4.9. *If $f \in \mathbb{F}_p[T]$ is monic and reducible, and is **not** of the form $\pi_1\pi_2$, where π_1 and π_2 are distinct monic irreducibles of the same degree, then the proportion of Miller–Rabin witnesses for f is at least 75%.*

This is analogous to the 75% lower bound for the proportion of Miller–Rabin witnesses in $\mathbb{Z}$. The monic reducible f that are not covered by Theorem 4.9 will be discussed in Theorem 4.12. Such exceptions to the theorem are Carmichael polynomials with two irreducible factors, which have no analogue in $\mathbb{Z}$ since Carmichael numbers have at least three prime factors.

Proof. We will count the number of Miller–Rabin-nonwitnesses for f and show

$$\frac{|\{\text{Miller–Rabin nonwitnesses } a \text{ for } f\}|}{\mathrm{N}(f)-1} \leq \frac{1}{4}.$$

We split the proof of this into two cases: (1) when f has an irreducible factor π such that $\pi^2 \mid f$, and (2) when f is squarefree.

Case 1: there exists an irreducible polynomial π in $\mathbb{F}_p[T]$ such that $\pi^2 \mid f$. Recall from right after Example 4.4 that every Miller–Rabin nonwitness for f is a Fermat nonwitness for f. Thus

$$|\{\text{Miller–Rabin nonwitnesses } a \text{ for } f\}| \leq |\{a^{\mathrm{N}(f)-1} \equiv 1 \bmod f \mid \deg a < \deg f\}|.$$

We will show the set of Fermat nonwitnesses for f has proportion less than 25%—i.e., $|\{a^{\mathrm{N}(f)-1} \equiv 1 \bmod f \mid \deg a < \deg f\}| < (\mathrm{N}(f)-1)/4$. Partition the Fermat nonwitnesses for f according to their reduction modulo π, which makes the number of Fermat nonwitnesses for f equal to the following sum:

$$\sum_{\substack{\deg b < \deg \pi \\ b \neq 0}} |\{a^{\mathrm{N}(f)-1} \equiv 1 \bmod f, \deg a < \deg f, a \equiv b \bmod \pi\}|.$$

Our goal is to show this sum is less than or equal to $(\mathrm{N}(f)-1)/4$.

By Lemma 4.7,

$$a^{\mathrm{N}(f)-1} \equiv 1 \bmod f, a \equiv b \bmod \pi \implies a \equiv b^{\mathrm{N}(f)} \bmod \pi^2,$$

so

$$\sum_{\substack{\deg b<\deg \pi,\\ b\neq 0}} |\{a^{\mathrm{N}(f)-1} \equiv 1 \bmod f, a \equiv b \bmod \pi\}| \;\leq \sum_{\substack{\deg b<\deg \pi,\\ b\neq 0}} |\{a \equiv b^{\mathrm{N}(f)} \bmod \pi^2\}|,$$

where the a's being counted on both sides have degree less than $\deg f$.

The set of nonzero polynomials with degree less than $\deg f$ has size $\mathrm{N}(f)-1$, and among such polynomials a the subset $\{a \equiv b^{\mathrm{N}(f)} \bmod \pi^2\}$ when $b \not\equiv 0 \bmod \pi$ has size $\mathrm{N}(f)/\mathrm{N}(\pi^2)$ since $\pi^2 \mid f$.

Therefore,

$$\begin{aligned}
\sum_{\substack{\deg b<\deg \pi,\\ b\neq 0}} |\{a \equiv b^{\mathrm{N}(f)} \bmod \pi^2, \deg a < \deg f\}| &= \sum_{\substack{\deg b<\deg \pi,\\ b\neq 0}} \frac{\mathrm{N}(f)}{\mathrm{N}(\pi^2)} \\
&= \frac{(\mathrm{N}(\pi)-1)\mathrm{N}(f)}{\mathrm{N}(\pi^2)} \\
&= \frac{(\mathrm{N}(\pi)-1)\mathrm{N}(f)}{\mathrm{N}(\pi)^2} \\
&= \frac{(\mathrm{N}(\pi)^2-1)\mathrm{N}(f)}{\mathrm{N}(\pi)^2(\mathrm{N}(\pi)+1)} \\
&\leq \frac{\mathrm{N}(f)(\mathrm{N}(\pi)^2-1)}{4\mathrm{N}(\pi)^2} \text{ since } \mathrm{N}(\pi) \geq 3 \\
&= \frac{\mathrm{N}(f) - \mathrm{N}(f)/\mathrm{N}(\pi)^2}{4} \\
&\leq \frac{\mathrm{N}(f)-1}{4}
\end{aligned}$$

since $\mathrm{N}(\pi)^2 \leq \mathrm{N}(f)$. Thus the proportion of Fermat nonwitnesses for f is at most $1/4$, so we get the same bound for the proportion of Miller–Rabin nonwitnesses for f, which means the proportion of Miller–Rabin witnesses for f is at least $3/4$.

Case 2: f is squarefree. Write $f = \pi_1\pi_2\ldots\pi_r$ for distinct monic irreducibles π_i and $r \geq 2$.

The number of a that are Miller–Rabin nonwitnesses for f is

$$|\{a^k \equiv 1 \bmod f\}| + \sum_{i=0}^{e-1} |\{a^{2^i k} \equiv -1 \bmod f\}|,$$

where the a's counted in each term have $\deg a < \deg f$. The Chinese remainder theorem gives this value as

$$\prod_{j=1}^{r} |\{a^k \equiv 1 \bmod \pi_j \mid a \in \mathbb{F}_p[T]^\times_{\pi_j}\}| + \sum_{i=0}^{e-1} \prod_{j=1}^{r} |\{a^{2^i k} \equiv -1 \bmod \pi_j \mid a \in \mathbb{F}_p[T]^\times_{\pi_j}\}|.$$

To count the first term, since $\mathbb{F}_p[T]_{\pi_j}^\times$ has size $\mathrm{N}(\pi_j)-1$ we have

$$\{a^k \equiv 1 \bmod \pi_j \mid a \in \mathbb{F}_p[T]_{\pi_j}^\times\} = \{a^{(k,\mathrm{N}(\pi_j)-1)} \equiv 1 \bmod \pi_j \mid a \in \mathbb{F}_p[T]_{\pi_j}^\times\},$$

which has $d_j := (k, \mathrm{N}(\pi_j)-1)$ elements since $\mathbb{F}_p[T]_{\pi_j}^\times$ is cyclic. Therefore

$$\prod_{j=1}^{r} |\{a^k \equiv 1 \bmod \pi_j \mid a \in \mathbb{F}_p[T]_{\pi_j}^\times\}| = \prod_{j=1}^{r} d_j.$$

Next, we count for each i from 0 to $e-1$

$$\prod_{j=1}^{r} |\{a^{2^i k} \equiv -1 \bmod \pi_j \mid a \in \mathbb{F}_p[T]_{\pi_j}^\times\}|.$$

By Lemma 4.8 applied to $x = a^{2^i}$, this product is

$$\prod_{j=1}^{r} |\{a^{2^i d_j} \equiv -1 \bmod \pi_j \mid a \in \mathbb{F}_p[T]_{\pi_j}^\times\}|.$$

To count the number of solutions in $\mathbb{F}_p[T]_{\pi_j}$ to $a^{2^i d_j} \equiv -1 \bmod \pi_j$, let g_j be a generator of $\mathbb{F}_p[T]_{\pi_j}^\times$. Then

$$|\{a^{2^i d_j} \equiv -1 \bmod \pi_j \mid a \in \mathbb{F}_p[T]_{\pi_j}^\times\}| = |\{g_j^{\alpha \cdot 2^i d_j} \equiv -1 \bmod \pi_j \mid 1 \le \alpha \le \mathrm{N}(\pi_j)-1\}|,$$

which is in turn equal to the number of solutions α to

$$\alpha \cdot 2^i d_j \equiv \frac{\mathrm{N}(\pi_j)-1}{2} \bmod \mathrm{N}(\pi_j)-1, \text{ where } 1 \le \alpha \le \mathrm{N}(\pi_j)-1.$$

This is the same as counting α such that $\alpha \cdot 2^{i+1} d_j$ is an odd multiple of $\mathrm{N}(\pi_j)-1$.

Note that d_j is odd since $d_j \mid k$, and $d_j \mid \mathrm{N}(\pi_j)-1$ by the definition of d_j. We also require $2^{i+1} \mid \mathrm{N}(\pi_j)-1$. Since $(2^{i+1}, d_j) = 1$, α must be an odd multiple of $\frac{\mathrm{N}(\pi_j)-1}{2^{i+1} d_j}$. There are exactly $2^i d_j$ such α in $\mathbb{Z}^+$ up to $\mathrm{N}(\pi_j)-1$ if $2^{i+1} \mid \mathrm{N}(\pi_j)-1$, and none otherwise.

Let 2^h be the highest power of 2 dividing all $\mathrm{N}(\pi_j)-1$. Then $2^h \mid \mathrm{N}(f)-1$, so $h \le e$ and

$$\begin{aligned}
\sum_{i=0}^{e-1} \prod_{j=1}^{r} |\{a^{2^i k} \equiv -1 \bmod \pi_j \mid a \in \mathbb{F}_p[T]_{\pi_j}^\times\}| &= \sum_{i=0}^{h-1} \prod_{j=1}^{r} 2^i d_j \\
&= \sum_{i=0}^{h-1} 2^{ir} \prod_{j=1}^{r} d_j \\
&= \prod_{j=1}^{r} d_j \cdot \Big(\frac{2^{hr}-1}{2^r-1}\Big).
\end{aligned}$$

Thus the total number of Miller–Rabin nonwitnesses for f is

$$\prod_{j=1}^{r} d_j + \prod_{j=1}^{r} d_j \cdot \Big(\frac{2^{hr}-1}{2^r-1}\Big) = \prod_{j=1}^{r} d_j \cdot \Big(1 + \frac{2^{hr}-1}{2^r-1}\Big).$$

For all π_j, we have $2^h d_j \mid \mathrm{N}(\pi_j) - 1$, so

$$\prod_{j=1}^{r} d_j \leq \frac{\prod_{j=1}^{r}(\mathrm{N}(\pi_j)-1)}{2^{hr}}.$$

In the numerator on the right, $\prod_{j=1}^{r}(\mathrm{N}(\pi_j)-1) < \prod_{j=1}^{r}\mathrm{N}(\pi_j) = \mathrm{N}(f)$ and these are integers, so $\prod_{j=1}^{r}(\mathrm{N}(\pi_j)-1) \leq \mathrm{N}(f)-1$. Therefore, the number of Miller–Rabin nonwitnesses for f is at most

$$\frac{\mathrm{N}(f)-1}{2^{hr}}\Big(1 + \frac{2^{hr}-1}{2^r-1}\Big).$$

Thus it suffices to show that

$$\frac{1 + \frac{2^{hr}-1}{2^r-1}}{2^{hr}} \leq \frac{1}{4}. \tag{7}$$

At this point, we prove a lemma.

Lemma 4.10. *If $f = \pi_1\pi_2$ where π_1 and π_2 are monic irreducible polynomials in $\mathbb{F}_p[T]$ of **different degrees**, then*

$$d_1 d_2 < \frac{\mathrm{N}(f)-1}{4^h \cdot 2},$$

where 2^h is the highest power of 2 dividing both $\mathrm{N}(\pi_1)-1$ and $\mathrm{N}(\pi_2)-1$, $\mathrm{N}(f)-1 = 2^e k$ with $e \geq 1$ and k odd, $d_1 = (k, \mathrm{N}(\pi_1)-1)$, and $d_2 = (k, \mathrm{N}(\pi_2)-1)$.

Proof. Since $\mathrm{N}(f) = \mathrm{N}(\pi_1)\mathrm{N}(\pi_2)$ where $\mathrm{N}(\pi_1)$ and $\mathrm{N}(\pi_2)$ are greater than 1, $\mathrm{N}(f)-1 > (\mathrm{N}(\pi_1)-1)(\mathrm{N}(\pi_2)-1)$ by some algebra.

Let 2^{h_1} and 2^{h_2} be the highest powers of 2 dividing $\mathrm{N}(\pi_1)-1$ and $\mathrm{N}(\pi_2)-1$ respectively, so $h = \min(h_1, h_2) \geq 1$, $2^{h_1} d_1 \mid \mathrm{N}(\pi_1)-1$, and $2^{h_2} d_2 \mid \mathrm{N}(\pi_2)-1$. Suppose $2^{h_1} d_1 \neq \mathrm{N}(\pi_1)-1$. Then $\mathrm{N}(\pi_1)-1 \geq 2^{h_1}(3d_1)$, so

$$\mathrm{N}(f)-1 > (\mathrm{N}(\pi_1)-1)(\mathrm{N}(\pi_2)-1) \geq 2^{h_1}(3d_1)\cdot 2^{h_2} d_2 \geq 4^h d_1 d_2 \cdot 3 > 4^h d_1 d_2 \cdot 2,$$

so the desired inequality holds. The same conclusion can be made if $2^{h_2} d_2 \neq \mathrm{N}(\pi_2)-1$. Therefore we can assume $\mathrm{N}(\pi_1)-1 = 2^{h_1} d_1$ and $\mathrm{N}(\pi_2)-1 = 2^{h_2} d_2$. From $\mathrm{N}(f) = \mathrm{N}(\pi_1)\mathrm{N}(\pi_2)$ it follows that d_1 and d_2 divide each other, so $d_1 = d_2$. Thus $h_1 \neq h_2$, since otherwise $\mathrm{N}(\pi_1) = \mathrm{N}(\pi_2)$, but π_1 and π_2 have different degrees and thus different norms. We have

$$\mathrm{N}(f)-1 > (\mathrm{N}(\pi_1)-1)(\mathrm{N}(\pi_2)-1) = 2^{h_1} d_1 \cdot 2^{h_2} d_2 \geq 2 \cdot 4^h d_1 d_2,$$

where the last inequality holds since $h_1 \neq h_2$. □

Now we can finish the proof of Case 2. The left side of (7) is

$$\frac{1}{2^{hr}}\Big(1-\frac{1}{2^r-1}\Big)+\frac{1}{2^r-1}\le\frac{1}{2^r}\Big(1-\frac{1}{2^r-1}\Big)+\frac{1}{2^r-1}=\frac{1}{2^r}+\frac{1}{2^r-1}\Big(1-\frac{1}{2^r}\Big),$$

which is at most $1/4$ for $r\ge 3$. If $r=2$, then by Lemma 4.10

$$\begin{aligned}\prod_{j=1}^{r} d_j\cdot\Big(1+\frac{2^{hr}-1}{2^r-1}\Big)&=d_1d_2\Big(1+\frac{4^h-1}{3}\Big)\\&<\frac{\mathrm{N}(f)-1}{4^h\cdot 2}\Big(1+\frac{4^h-1}{3}\Big)\\&=\big(\mathrm{N}(f)-1\big)\frac{4^h+2}{6\cdot 4^h}\\&\le\big(\mathrm{N}(f)-1\big)\left(\frac{1}{6}+\frac{2}{6\cdot 4}\right)\\&=\frac{\mathrm{N}(f)-1}{4},\end{aligned}$$

completing the proof of Theorem 4.9. □

Next we will estimate the proportion of Miller–Rabin witnesses for the monic reducible f in $\mathbb{F}_p[T]$ that were not accounted for in Theorem 4.9: $f=\pi_1\pi_2$ where the π_j's are monic irreducibles with the same degree. The argument will use the following lemma.

Lemma 4.11. *For $v,\ell\in\mathbb{Z}^+$,*

$$\frac{4^v\ell+2\ell}{4^v\ell+2^{v+1}}\in[3/4,3/2).$$

Proof. The inequality

$$\frac{3}{4}\le\frac{4^v\ell+2\ell}{4^v\ell+2^{v+1}}$$

is equivalent to $(4^v+8)\ell-6\cdot 2^v\ge 0$, and

$$(4^v+8)\ell-6\cdot 2^v\ge 4^v+8-6\cdot 2^v=(2^v-6)2^v+8,$$

which is positive when $v\ge 3$. At $v=1$ and 2, the right side is 0.

The other inequality

$$\frac{4^v\ell+2\ell}{4^v\ell+2^{v+1}}<\frac{3}{2}$$

is equivalent to $(4^v-4)\ell+6\cdot 2^v>0$, and

$$(4^v-4)\ell+6\cdot 2^v\ge 4^v-4+6\cdot 2^v\ge 6\cdot 2^v>0.$$ □

Theorem 4.12. *If $f\in\mathbb{F}_p[T]$ is of the form $\pi_1\pi_2$, where π_1 and π_2 are distinct monic irreducibles with equal degree, then the proportion of Miller–Rabin witnesses for f lies in the interval $(1/2,3/4]$.*

Proof. The formula from the proof of Theorem 4.9 for counting Miller–Rabin witnesses for f,

$$\prod_{j=1}^{r} d_j \cdot \Big(1 + \frac{2^{hr} - 1}{2^r - 1}\Big),$$

still holds in this case. Let $\deg \pi_1 = \deg \pi_2 = D$, so $\mathrm{N}(\pi_1) - 1 = \mathrm{N}(\pi_2) - 1 = p^D - 1$. Write $p^D - 1 = 2^v \ell$ for $v \geq 1$ and odd $\ell \geq 1$. We see that f has degree $2D$, making

$$\mathrm{N}(f) - 1 = p^{2D} - 1 = (2^v \ell)(2^v \ell + 2) = 2^{v+1} \ell (2^{v-1} \ell + 1).$$

Since ℓ is odd, $\mathrm{N}(f) - 1 = 2^e k$ where $e \geq v + 1$ and $k = \ell m$, where m is the odd part of $2^{v-1}\ell + 1$ (which is $2^{v-1}\ell + 1$ unless $v = 1$). Thus

$$d_1 = (k, \mathrm{N}(\pi_1) - 1) = (\ell m, 2^v \ell) = \ell,$$

and likewise $d_2 = \ell$.

Plugging these values into the formula above, the number of Miller–Rabin witnesses for f is

$$\ell^2 \Big(1 + \frac{4^v - 1}{3}\Big).$$

Thus the proportion of Miller–Rabin nonwitnesses for f is

$$\frac{\ell^2 \cdot (4^v + 2)/3}{\mathrm{N}(f) - 1} = \frac{\ell^2 \cdot (4^v + 2)/3}{2^{v+1} \ell (2^{v-1} \ell + 1)} = \frac{\ell (4^v + 2)}{3 \cdot 2^{v+1} (2^{v-1} \ell + 1)} = \frac{1}{3} \cdot \frac{4^v \ell + 2\ell}{4^v \ell + 2^{v+1}},$$

which by Lemma 4.11 lies in $[1/4, 1/2)$, so the proportion of Miller–Rabin witnesses for f lies in $(1/2, 3/4]$. □

Remark 4.13. This exceptional case occurs because in Lemma 4.11, the inequality $2^{h_1} d_1 \cdot 2^{h_2} d_2 \geq 2 \cdot 4^h d_1 d_2$ does not generally hold if π_1 and π_2 have the same degree.

5 Conclusion

In this paper, we developed polynomial analogues of the Fermat test, Solovay–Strassen test, and Miller–Rabin test from the integers to the ring $\mathbb{F}_p[T]$, with p odd in the third test. For the Solovay–Strassen test, we addressed both odd and even characteristic cases, introducing a novel additive irreducibility test in $\mathbb{F}[T]$ when $\mathbb{F}$ is any finite field of characteristic 2. These tests can be carried out efficiently. In the cases of the Solovay–Strassen and Miller–Rabin tests, there is nothing analogous to Carmichael numbers, which means these two tests allow us to probabilistically detect if a polynomial is irreducible without having to factor it. These results not only highlight intriguing parallels between integer primality testing and polynomial irreducibility testing, but also have meaningful implications in computational number theory, cryptography, and Galois theory.

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